NATIONAL INSTITUTE OF TECHNOLOGY JAMSHEDPUR

B. Tech. in

Production and Industrial Engineering

SYLLABUS FOR NEP-BASED CURRICULUM

for B. Tech. Program

(Effective from 2021-22)

NATIONAL INSTITUTE OF TECHNOLOGY JAMSHEDPUR

VISION

To be one of the premier technical institutions for its academic excellence and innovative research to meet the future needs of the society.

MISSION

M1: To build conducive environment for learning and creativity.

M2: To train students to become technically competent professionals and socially responsible citizens.

M3: To develop innovative products and technologies for the betterment of the society.

DEPARTMENT OF PRODUCTION AND INDUSTRIAL ENGINEERING

Brief about the Department:

Production and Industrial Engineering was started since 1988 with the name of Production Engineering & Management Department. The name of the department has been changed to Production & Industrial Engineering with effect from 2007. The Department developed different laboratories like CAD/CAM Lab., Welding Lab., Foundry lab., and Robotics lab. Development of two other laboratories namely “Operations and Supply chain Lab.” and “Non-traditional Manufacturing Lab.” are in process. Time to time curricula/syllabus is upgraded as per the market and technology requirement. Along with the undergraduate program, the department is running one post graduate program called “Manufacturing Systems Engineering”. The department is also running Ph.D. programs. The department is also planning to start a few more post graduate programs namely “Production and Industrial Engineering” and “Operations and Supply Chain Management”.

List of Programs Offered by the Department

Program	Specialization
B. Tech.	Production and Industrial Engineering
M. Tech	Manufacturing Systems Engineering
Ph.D.	Production and Industrial Engineering

VISION

To nurture mechanical engineers with passion for professional excellence, ready to take global challenges and to serve the society with high human values.

MISSION

M1: To educate the students (B.Tech. M. Tech. and Ph.D.) with fundamental knowledge in concerned subjects, and to update them with the state-of-the-art in the field.

M2: To develop and maintain laboratories which create to the needs of students (B.Tech. M.Tech. and Ph.D.).

M3: Quest for inter-disciplinary research.

M4: To promote academia-industry collaboration.

Program Educational Objectives (PEOs)

PEO1: To create a congenial environment that promotes learning, growth and imparts the ability to work with multi-disciplinary groups in professional, industry and research organizations.

PEO2: To prepare students to be able to apply concepts of mathematics, science and computing to Production and Industrial Engineering.

PEO3: To prepare students to be employed in jobs related to designing, modeling, analyzing, and managing modern complex systems in manufacturing and service sectors at local, regional, national and global levels.

PEO4: To broaden and deepen their capabilities in analytical and experimental research methods, analysis of data and drawing relevant conclusions for scholarly writing and presentation.

Consistency of PEOs with Mission of the Department

	M1	M2	M3	M4
PEO1	3	3	3	2
PEO2	3	3	3	2
PEO3	3	3	3	3
PEO4	2	3	3	3

Correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) **2: Moderate (Medium)** **3: Substantial (High)**

PROGRAM OUTCOMES (POs)

Engineering Graduates will be able to:

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one’s own work, as member and leader in a team, to manage projects and multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

Graduates of the program will be able to

PSO1: Design, analyze and handle the modern manufacturing systems.

PSO2: Construct and analyze the problems related to Production and Industrial

Engineering by using modern tools.

B. Tech. in Production and Industrial Engineering Based on NEP (PROGRAM ARTICULATION MATRIX)

Semester	Course Code	Course Name	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	
	PI1301	Manufacturing Process-I	3	2.7	2.7	2.7	2.5	1	1					2.7	2.7		
	PI1302	Measurement and Metrology	3	3	1.5	2	2				1	1		2	2	2.5	
	ME1301	Mechanics of Solids	3	3	2.3	2.3		2		1	1			3	3		
	ME1302	Engineering Thermodynamics	3	3	3	2.8		2	2					3	3		
	ME1304	Theory of Machines	3	3	3	2.8	2	2		1	1			3	3		
	PI1303	Indian Art, Culture and Science/Vedic Mathema	3	3	3	2		2	2		2	2	2	3	3	3	
	PI1304	Manufacturing Process-I Lab.	3	3	3	2		2	2					3	3		
3rd	PI1305	Measurement and Metrology Lab.	3	3	3	2		2	2					3	3		
	PI1401	Manufacturing Process-II	3	2.7	2.5	2.5	2.5	1	1					2.5	2		
	PI1402	Engineering Economics and Accountancy	2.5	2	2.5	2.3	2.8	0.3	0.3	1	0.8	0.25	3	2.5	1.5	2.5	MINOR
	PI1403	Production Management	3	2	2.3	1.8	2.3	2	0.8	1.3	1	1.75	2.75	2.5	2.25	3	
	ME1409	Design of Machine Elements	3	3	3	2.8	2	2		1	1			3	3		
	ME1410	Fluid Mechanics and Machinery	3	3	3	2.8		2	2					3	3		
	PI1404	Manufacturing Process-II Laboratory	3	3	3	2		2	2					3	3		
4th	ME14XX	Fluid Mechanics and Machinery Lab.	3	3	3	2		2	2					3	3		
	PI1501	CAD/CAM	2.8	3	2.5	2	2.5							2	2	1.5	
	PI1502	Supply Chain Management	3	2	2.3	1.8	2.3	2	0.8	1.3	1	0.25	2.75	2.5	2.25	3	MINOR
	PI1503	Product Design and Development	3	3	3	2.8	2	2		1	1			3	3		
	PI1504	Operations Research	3	2	2.3	1.8	2.3	2	0.8	1.3	1	0.25	2.75	2.5	2.25	3	MINOR
	PI1505	Surface Engineering	3	2.7	3	2.7	2	1	1					2.7	2.7		
	PI1506	Precision Engineering	3	2.5	2.5	1.8	2.8	1		2.3				3	3		
	PI1507	Strategic Management	2	2.3	2.8	1.5	1.5	2.5	1	1.3	1	1	2.75	2.5		3	
	PI1508	Digital Manufacturing	3	2.7	2.5	2.7	2.5	1	1	1				2.7	2.5	1	
5th	HSIR13	Fundamentals of Intellectual Property Right						1	1	1.25	1	1	2.75	2.5	2	2	
	PI1601	Non-traditional Manufacturing Processes	2	2.5	2	2.5	3	2	2.5		1	2		3	3	2	
	PI1602	Metal Forming	3	2.7	3	2.7	2.5	1	1					2.5	2.5	0.25	MINOR
	PI1603	Design of Production Tooling	3	2.3	2.3	1.8	2	2	0.5	2	2.3	2	1.5	2.5		3	
	PI1604	Energy Conservation and Management	3	2.7	2.5	2.7	2.5	1	1	1				2.7	2.5	1	
	PI1605	Concurrent Engineering	2	2.5	1.8	2.3		2	1	1.8	0.8	2.25	2.5	3	2.75	2.75	
	PI1606	Micro and Nano Manufacturing	3	2.3	2	1.8	2.8	2	0.3	2				2.75	3		
	PI1607	Project Management	3	3	3	2.8	3	1.5	2	0.3	2.3	2.5	3	2.25	0.75	3	
	PI1608	Advanced Materials	3	3	2.8	1.3	2.3	1.3	0.7	2				3	2.5		
	PI1609	Reliability Engineering	1.5	0.5	0	1	1	0	0	0	1	3	1.5	3	2.5	1.75	
	PI1610	Work System Design and Ergonomics	1.8	2.8	1.3	2	2.3	2	0.8	1.5	2.5	2.75	2.25	2	2.5	2.75	
	PI1611	Near Net-shape Manufacturing	3	2.7	2.5	2.7	2.5	1	1	1				2.7	2.5	1	
	HS1606	Soft Skills								2	3	3	3	3	1	2	
	PI1612	Management Information Systems						3	2	2		3	1.75	2.75	1.75	2.25	
	PI1613	Total Quality Management	3	3	2.3	1.8	1.8	3	2	1.3	2.5	3	1.75	2.75	1.75	2.25	MINOR Open Elective-I
	PI1614	Business Statistics	3	3	3	3	2.3	0.3			1.3	2	2	2	1	2.5	
6th	PI1615	Additive Manufacturing	3	3	2.7	2.7	2.5	1	1	1	1.3			3	2.7	1	
	PI1701	Automation and Robotics	3	2.7	2.7	3	2.7	1	1					3	2.7	1	
	PI1702	Introduction to Machine Learning	3	2.7	2.7	3	2.7	1			1		3	3		2.7	
	PI1703	Characterization of Engineering Materials	3	2.5	1.5	0.7	2.5		1	2.3				3	3		
	PI 1704	Design for Manufacturing and Assembly	3	3	3	2.8		2		1	1			3	3	2	
	PI1705	Value Engineering	2.5	3	1.5	1.8	1.3	1.5	1	1.3	1.8	2	2	2.5	2	2	
	PI1706	Modeling and Simulation	2.3	2.8	2.5	3	2.5				2.5	2	3	2	2	3	
	PI1707	Finite Element Method	3	2.7	2.7	2.5	3							2.7	2.5	1	
	PI1708	Lean Manufacturing	2.5	2.8	2.3	2.3	2.3	2.3	2.8	1	2.3	2.75	2.25	2.75	1.75	2.25	
	PI1709	Safety in Engineering Industry	2	2				3	2.3		3			2.7	2.5	1	
	PI1710	Integrated Computational Materials Engineering	3	3	2.7	3	3	1	1					2.5	2.7	2	
	PI1711	Sustainable Manufacturing and Impact Assessm	3	2.8	2.3	1.5	2	3	3	3		1.33		3		2.5	
	PI1712	Circular Economy	1.8	2.5	2.3	2.3	1	2	3	1	2.8	2.25	2.25	2			MINOR Open Elective-II
	PI1713	Industry 4.0	3	3	2.8	2.8	3	1.3	2	0.3	2.5	0.75	0.5	2	3	3	
	PI1714	Warehouse Management	3	2	2	1.8	1.3	1.5				1	2.75	2.5	1.5	3	
	PI1715	Automation and Robotics Lab.	3	3	3	2		2	2					3	3		
7th	PI1716	Industrial Training	3	3	3	2	2	2	2	1	1	2	2	3	3	2	
8th	PI1801	Project Work/ Internship	3	3	3	2	2	2	2	2	2	2	2	3	3	3	
		Average	2.81	2.68	2.49	2.23	2.27	1.67	1.42	1.32	1.57	1.82	2.31	2.70	2.47	2.17	

CURRICULUM

B.Tech (Production and Industrial Engineering)

Minor (Production and Industrial Engineering)

The total minimum credits required for completing the B. Tech. program is 160-170 for major degree and extra 20 credits for minor degree.

Minimum Credit Requirement for the Various Course Categories

Course Category	Courses	No. of Credits	Weightage (%)
Common Institute Requirement Courses (CIR)	24	54	33
Program Core Courses (PC)	19	54	33
Program Electives (PE)	05	15	9
Open Electives (OE)	07	20	12
Project Work/Internship (PW)	02	14	8
ONLINE Course (OC)	02	06	4
Extra Academic Activity Non-credit Course (NC)	02	02	1
Total	61	**165**	**100**
Minor (Optional)	07 Courses for 20 credits	20 Additional credits	

CIR Courses:

Sl. No.	Name of the Course	Number of Courses	Max. Credits
1.	Engineering Mathematics	02	08
2	Engineering Physics	01	03
	Engineering Physics Lab.	01	01
3	Engineering Chemistry	01	03
	Engineering Chemistry Lab.	01	01
4	Basic Electronics and Instrumentation	01	03
	Basic Electronics and Instrumentation Lab.	01	01
5	Electrical engineering and measurement	01	03
	Electrical engineering and measurement Lab.	01	01
6	Introduction to Programing and Data Structures	01	03
	Introduction to Programing and Data Structures Lab	01	01
7	English for Communication	01	02
	Language Lab.	01	01
8	Engineering Mechanics	01	04
	Engineering Mechanics Lab.	01	01
9	Environmental Science	01	02
10	Science of Living Systems	01	02
11	Material Science	01	03
12	Workshop Practices	01	02
13	Engineering Drawing and Computer Graphics	01	02
14	IPR	01	03
15	Soft Skill	01	02
16	Comprehensive Viva (Internship)	01	02
Total		**24**	**54**

I Year: I-Semester [1st Semester]

Sl. No.	Course Code	Course Name	L	T	P	Credits
1	PH1101	Engineering Physics	3	0	0	3
2	MA1101	Engineering Mathematics-I	3	1	0	4
3	EC1101	Electronics and Instrumentation	3	0	0	3
4	ME1101	Engineering Mechanics	3	1	0	4
5	HS1101	English for Communication	3	0	0	3
6	HS1102	Science of Living Systems	2	0	0	2
7	MF1101	Workshop Practices	0	0	3	2
8	EC1102	Electronics and Instrumentation Lab.	0	0	2	1
9	ME1102	Engineering Mechanics Lab.	0	0	2	1
10	PH1102	Engineering Physics Lab.	0	0	2	1
11	HS1103	Yoga/NSS/NCC/Life Skills				
		Total	17	2	9	24

I Year: II-Semester [2nd Semester]

Sl. No.	Course Code	Course Name	L	T	P	Credits
1	CH1201	Engineering Chemistry	3	0	0	3
2	MA1202	Engineering Mathematics-II	3	1	0	4
3	EE1201	Electrical Engineering and Measurement	3	0	0	3
4	CS1201	Introduction to Programming and Data Structures	3	0	0	3
5	MM1201	Material Science	3	0	0	3
6	CE1201	Environmental Science	2	0	0	2
7	ME1203	Engineering Drawing and Computer Graphics	0	0	3	2
8	EE1202	Electrical Engineering and Measurement Lab.	0	0	2	1
9	CS1202	Introduction to Programming and Data Structures Lab.	0	0	2	1
10	CH1202	Engineering Chemistry Lab.	0	0	2	1
		Total	17	1	9	23

II Year: I-Semester [3rd Semester]

Sl. No.	Course Code	Course Name	L	T	P	Credits
1	PI1301	Manufacturing Process-I	3	1	0	4
2	PI1302	Measurement and Metrology	3	0	0	3
3	ME1301	Mechanics of Solids	3	0	0	3
4	ME1302	Engineering Thermodynamics	3	0	0	3
5	ME1304	Theory of Machines	3	0	0	3
6	PI1303	Indian Art, Culture and Science/Vedic Mathematics	2	0	0	2
7	PI1304	Manufacturing Process-I Lab.	0	0	2	1
8	PI1305	Measurement and Metrology Lab.	0	0	2	1
		Total	17	1	4	20

II Year: II-Semester [4th semester]

Sl. No.	Course Code	Course Name	L	T	P	Credits
1	PI1401	Manufacturing Process-II	3	1	0	4
2	PI1402	Engineering Economics and Accountancy	3	0	0	3
3	PI1403	Production Management	3	0	0	3
4	ME1409	Design of Machine Elements	3	1	0	4
5	ME1410	Fluid Mechanics and Machinery	3	0	0	3
6	PI1404	Manufacturing Process-II Laboratory	0	0	2	1
7	ME14XX	Fluid Mechanics and Machinery Lab.	0	0	2	1
		Total	15	2	4	19

III Year: I-Semester [5th semester]

Sl. No.	Course Code	Course Name	L	T	P	Credits
1	PI1501	CAD/CAM	3	0	0	3
2	PI1502	Supply Chain Management	3	0	0	3
3	PI1503	Product Design and Development	3	0	0	3
4	PI1504	Operations Research	3	1	0	4
5	PI15XX	Professional Elective-I	3	0	0	3
6	HS1502	Fundamentals of Intellectual Property Right	3	0	0	3
7	PI1509	CAD/CAM Laboratory	0	0	2	1
8	PI1510	Operations and Supply Chain Lab.	0	0	2	1
		Total	18	1	4	21

Professional Elective-I

PI1505 - Surface Engineering
PI1506 - Precision Engineering
PI1507 - Strategic Management
PI1508 - Digital Manufacturing

III Year: II-Semester [6th semester]

Sl. No.	Course Code	Course Name	L	T	P	Credits
1.	PI1601	Non-traditional Manufacturing Processes	3	0	0	3
2.	PI1602	Metal Forming	3	0	0	3
3.	PI1603	Design of Production Tooling	3	1	0	4
4.	PI16XX	Professional Elective-II	3	0	0	3
5.	PI16XX	Professional Elective-III	3	0	0	3
6.	PI16XX	Open Elective-I	3	0	0	3
7.	HS1606	Soft Skills	2	0	0	2
8.	PI1616	Non-traditional Manufacturing Processes Lab.	0	0	2	1
		Total	20	1	2	22

Professional Elective-II & III

PI1604 - Energy Conservation and Management
PI1605 - Concurrent Engineering
PI1606 - Micro and Nano Manufacturing
PI1607 - Project Management
PI1608 - Work System Design and Ergonomics
PI1609 - Advanced Materials
PI1610 - Reliability Engineering
PI1611 - Near Net-shape Manufacturing

Open Elective-I

PI1612 - Management Information Systems
PI1613 - Total Quality Management
PI1614 - Business Statistics
PI1615 - Additive Manufacturing

IV Year: I-Semester [7th Semester]

Sl. No.	Course Code	Course Name	L	T	P	Credits
1	PI1701	Automation and Robotics	3	0	0	3
2	PI1702	Introduction to Machine Learning	3	0	0	3
3	PI17XX	Professional Elective-IV	3	0	0	3
3	PI17XX	Professional Elective-V	3	0	0	3
4	PI17XX	Open Elective-II	3	0	0	3
5	PI1715	Automation and Robotics Lab.	0	0	2	1
7	PI1716	Industrial Training				2
		Total	15	0	2	18

Professional Elective-IV & V

PI1703 - Characterization of Engineering Materials

PI1704 - Design for Manufacturing and Assembly

PI1705 - Value Engineering

PI1706 - Modeling and Simulation

PI1707 - Finite Element Method

PI1708 - Lean Manufacturing

PI1709 - Safety in Engineering Industry

PI1710 - Integrated Computational Materials Engineering

Open Elective-II

PI1711 - Sustainable Manufacturing and Impact Assessment

PI1712 - Circular Economy

PI1713 - Industry 4.0

PI1714 - Warehouse Management

IV-Year: II-Semester [8th Semester]

Sl. No.	Course Code	Course Name	L	T	P	Credits
1	PI1801	Project Work/ Internship				12
	PI1802	MOOC1	3	0	0	3
	PI1803	MOOC2	3	0	0	3
		Total				18

B. Tech. in Production and Industrial Engineering: **165 Credits**

Minor in Production and Industrial Engineering: **20 Credits**

- ❖ Those who wish to go for additional (minor) specialization must take additional courses of a total 18-20 credits during 4th to 7th semester (evenly distributed throughout the semesters).
- ❖ Students from all the departments can opt for Minor in Production and Industrial Engineering.

Sl. No.	Course Code	Course Name	L	T	P	Credits
1	PI1402	Engineering Economics and Accountancy	3	0	0	3
2	PI1502	Supply Chain Management	3	0	0	3
3	PI1504	Operations Research	3	1	0	4
4	PI1602	Metal Forming	3	0	0	3
5	PI16XX	Open Elective-I	3	0	0	3
6	PI17XX	Open Elective-II	3	0	0	3
7	PI1510	Operations and Supply Chain Lab.	0	0	2	1
		Total	18	1	2	20

- ❖ B. Tech. Student scoring CGPA greater than or equal to 8.0 will be awarded B. Tech. (honors).

II Year: I-Semester [3rd Semester]

PI1301: Manufacturing Process-I **(3-1-0)**

Introduction to metal casting processes: Casting terms, Sand mold making procedure, Composition of molding sand and its different properties, Advantages and limitations of casting processes, Application of casting process.

Patterns, Materials, Allowances in patterns, Types of patterns, Patten color code, Gating and riser design, Special casting techniques - Shell mold casting, Investment casting, Die casting and its variants, Centrifugal casting and its variants, Continuous casting.

Melting practices: different types of melting furnaces-cupola, muffle, induction etc. and their working principles and applications, modern melting furnaces and techniques, Analysis of casting defects and their remedial measures, Automation in Foundry.

Introduction to Metal forming processes: Classification of metal forming processes, Plastic deformation and Yield criteria, Hot and Cold working, Forging, Rolling, Extrusion, Wire drawing, Sheet metal operations.

Classification of joining processes, Types of joints and welding positions, Weld nomenclature, Welding heat sources and their characteristics.

Various welding processes: Electric arc welding, Gas welding, Resistance welding, Solid state welding processes, non-conventional welding processes, Metallurgical characteristics of welded joints, Welding defects, Weld testing and Inspection.

Text and Reference Books:

1. R. K. Jain, 'Production Technology: Manufacturing Processes, Technology and Automation", 17th Edition
2. Ghosh and Malik, 'Manufacturing Science', TMH
3. P.C. Pandey, 'Production Engineering Science',
4. Kalpakjian, 'Manufacturing Engineering & Technology', Pearson
5. P. N. Rao, 'Manufacturing Technology', MCGRAW HILL INDIA
6. Lindberg, 'Manufacturing Processes', Pearson.
7. Kalpakjian, 'Manufacturing Processes for Engineering materials', Pearson
8. Kaushish, 'Manufacturing Processes' PHI
9. Jain, Principles of Foundry Technology, MCGRAW HILL INDIA

Course Outcomes: After completion of this course,

CO1: Students will design core, core print and gating system in metal casting processes.

CO2: Students will analyze different bulk and sheet metal forming processes for different engineering products.

CO3: Students will examine weld joints fabricated through solid state and fusion joining, brazing and soldering techniques.

CO4: Students will analyze the different joining processes and their applications.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	2	1	1					3	3	
CO2	3	3	3	3	3	1	1					3	3	
CO3	3	2	3	3	3	1	1					2	2	
CO4	3	3	2	3	2	1	1					3	3	
AVG	3	2.7	2.7	2.7	2.5	1.0	1.0					2.7	2.7	

PI1302: Measurement and Metrology **(3-0-0)**

Definitions, precision and accuracy, linearity, repeatability, sensitivity, readability, types and source of errors. Linear measurement, Vernier calipers, Vernier height master, micrometer, tests for their accuracies.

Limits, fits and gauges, definition and terminology, compound tolerances, tolerance accumulation, fundamental deviation, calculation of limits, clearance, tolerance etc., types of fits.

Comparator: characteristics, uses and special requirements, sigma mechanical comparators, reed type mechanical comparator, pneumatic comparator, electrical comparator, relative merits and demerits of comparators.

Measurement by light wave interference, flatness testing, fringe patterns, N.P.I. flatness interferometer, Coordinate measuring machine (CMM). Angular measurement sine center, sine bar, straightness test by sprit level, and auto collimator.

Measurement and testing of spur gears, terminology of gear tooth, involutes curve, rolling gear test, measurement of tooth thickness by constant chord method, gear tooth Vernier caliper, base tangent method, base pitch measurement, run-out measurement, tooth profile checking.

Measurement of surface finish, terminologies, center line, average root mean square, maximum peak to valley height, surface inspection of comparison methods, roughness measurement by instruments.

Measurement of screw thread, screw terminology, pitch errors in screw thread, measurement of major diameter, minor diameter and effective diameter, two wire method and three wire method.

Force measurement: Beam Balance, various types of loads

Torque measurement: Various types of dynamometers, characteristics of dynamometers, direct power measurement systems

Text and Reference Books:

1. R. K. Jain, ‘Engineering Metrology’, Khanna Publishers
2. O. P. Khanna, ‘Engineering Metrology’, Khanna Publishers
3. I. C. Gupta, ‘Text Book of Engineering Metrology’, Dhanpat Rai Publishing Company.
4. K. J. Hume, ‘Engineering Metrology’, Macdonald and Co., London, 2003.

5. Czichos, ‘Handbook of Metrology and Testing’, Springer.
6. Jay. L. Bucher, ‘The Metrology Hand book’, American Society for Quality, 2004
7. G. T. Smith, “Industrial Metrology”,Spinger, 2002.
8. J. W. Greve, F. W. Wilson, ‘Hand book of industrial metrology’ PHI – New Delhi
9. D. M. Anthony, “Engineering Metrology”, Pergamon Press.

Course Outcomes: After completion of this course,
CO1: Identify the fundamentals of various methods of measurements, metrology and measuring devices.
CO2: Apply the principle of linear and angular measuring instruments for the accurate and precise measurement of a given quantity
CO3: Develop competence in measuring devices with associated parameters
CO4: Use various devices for measuring torque, force, strain, stress and temperature.

CO-PO-PSO Mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1	2	2				1	1		2	2	2
CO2	3	3	2	2	2				1	1		2	2	3
CO3	3	3	2	2	2				1	1		2	2	3
CO4	3	3	1	2	2				1	1		2	2	2
AVG	3	3	1.5	2	2				1	1		2	2	2.5

ME1301: Mechanics of Solids **(3-0-0)**

Analysis of strain: Principal strain, Strain rosette, Mohr’s circle of strain & strain Rosette.
Fatigue: Fatigue of metal, Bauschinger’s experiment, strain method of obtaining fatigue ranges formula connecting stress range, Maximum stress and ultimate strength S-N curve, Gerber’s formula, Goodman’s law.

Strain Energy: Strain energy due to direct bending. Castigliano’s theorem, application to deflection of simply supported beams and cantilever beams due to shear.

Thick cylinders and spheres: thick cylinders, Radial and hoop stresses. Application of compound stress theories, compound cylinders, thick spherical- shell radial and circumferential stresses. Rotation of rings and discs: thin disc of uniform thickness, Radial and hoop stresses, Disc with central holes, Disc of uniform strength.

Theories of yielding: Different theories of failure, Comparison of theories of failure, yield loci.
Unsymmetrical bending: Flexural stresses due to unsymmetrical bending of beams.

Shear Centre: Shear center for thin-walled open cross flow section, shear flow.

Creep: Creep of metals, Mechanisms of creep; Equi-cohesive temperature; Creep curve, Creep rate; Prediction of long-term properties from short duration test.

Text and Reference Books:

1. George E. Dieter, ‘Mechanical Metallurgy’, McGraw Hill Education
2. Stephen Timoshenko and James M. Gere, ‘Mechanics of Materials’, CBS Publication.

3. Sadhu Singh, 'Strength of Materials', Khanna Publication
4. Richard W. Hertzberg, Richard P. Vinci, Jason L. Hertzberg, 'Deformation and Fracture Mechanics of Engineering Materials', Wiley

Course Outcomes: After completion of this course,

CO1: Students will describe the stress, strain and Mohr circle.

CO2: Students will solve problems related to deflection of beams.

CO3: Students will conduct the stress-strain analysis on thick and thin shells.

CO4: Students will solve problems related to theory of failure, bending, shear stress and creep.

CO-PO-PSO Mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1	1		2		1	1			3	3	
CO2	3	3	3	3		2		1	1			3	3	
CO3	3	3	3	2		2		1	1			3	3	
CO4	3	3	3	3		2		1	1			3	3	
AVG	3	3	2.25	2.25		2		1	1			3	3	

ME1302: Engineering Thermodynamics (3-0-0)

Basic concept: Dimensions and units, thermodynamic systems and their properties, zeroth low and temperature equilibrium concept. First law of thermodynamics: Concept of work and heat, open and closed systems, internal energy and enthalpy flow work example. Second law of thermodynamics: Introduction, Kelvin Planks and Clausius statements and their equivalence, reversibility, irreversibility, reversible cycle, Carnot cycle, corollaries of second law, Clausius inequality, entropy as a property, principle of increase of entropy.

Power Cycles: Rankine cycle – Ideal, Reheat and Regenerative Rankine cycles. Gas Power Cycles: Gas Power Cycles; Otto cycle, Diesel cycle, Dual cycle and Brayton cycle.

Refrigeration Cycles: Vapor compression refrigeration, Absorption refrigeration and Gas refrigeration Cycles.

Heat Transfer: Modes of heat transfer, conduction – Fourier law, thermal conductivity of solid, liquid and gases, factor influencing thermal conductivity, general 3-dimensional heat conduction equation. One-dimensional steady state conduction through plane walls, cylinder, spheres, flow through composite walls, cylinder and spheres, critical thickness of insulation, heat conduction with internal heat source through plane wall and cylinders. Two-dimensional steady state conduction through plane wall, one-dimensional unsteady state heat conduction, heat capacity systems.

Convection: Free and forced convection, basic concept of hydrodynamics thermal boundary layers, laminar and turbulent boundary layer over a flat plate, equation of motion and energy.

Radiation: Thermal radiation, monochromatic and total emissive power, absorptivity, reflectivity and transmissivity, black and gray bodies, Plank law, Wein law, Stefan Boltzman law, Kirchoff's law, heat transfer by radiation between black and gray surfaces.

Text and Reference Books:

1. P. K. Nag, ‘Engineering Thermodynamics’, Tata McGraw-Hill Education.
2. Yunus A. Cengel and Michael A. Boles, ‘Themodynamics: An Engineering Approach’, Tata McGraw-Hill Education.
3. Yunus A. Cengel, ‘Heat Transfer’, Tata McGraw-Hill Education.

Course Outcomes: After Completion of this course,
CO1: Students will describe the laws of thermodynamics and energy exchange processes in terms of various forms of energies like heat and work in the systems.
CO2: Students will analyse the power cycles.
CO3: Students will demonstrate the concepts of heat and mass transfer.
CO4: Students will solve problems related to heat and mass transfer.

CO-PO-PSO Mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	3		2	1					3	3	
CO2	3	3	3	3		2	1					3	3	
CO3	3	3	3	2		2	3					3	3	
CO4	3	3	3	3		2	3					3	3	
AVG	3	3	3	2.75		2	2					3	3	

ME1304: Theory of Machines **(3-0-0)**

Basic kinematics concepts: Links, kinematic pairs, mechanism and inversion.

Friction devices: Introduction, belt, chain and, rope drives, transmission of power through friction clutch, brakes.

Fundamental law of gearing and gear trains: Classification of gears, basic terminology, geometric and kinematic characteristics of involute and cycloidal tooth profiles, undercutting and interference. Simple, compound and planetary gear train, tooth load and torque.

Cams: Classification of cams and follower, radial cam, nomenclature, types of follower motions: uniform, simple harmonic, parallel cycloidal. Generation of cam profile by graphical method, pressure angle, cams with specific contours.

Vibration: Basic concept, SHM, degree of freedom, types of damping, equivalent system, free and forced vibration, linear and angular single degree freedom system with and without damping, whirling of shaft, vibration isolation and absorber, elementary treatment of system with two-degree of freedom.

Text and Reference Books:

1. G.H. Martin, ‘Kinematics and Dynamics of Machines’, 3rd edition, McGraw-Hill, 1982.
2. A. Ghosh, and A. K. Mallik, ‘Theory of Mechanisms and Machines’, 2nd edition, affiliated East-West Press., 2003.
3. T. Bevan, ‘Theory of Machines’, 3rd edition, CBS Publishers, 2004.
4. J. J. Vicker, J. E. Shigley, and G. R. Penock, ‘Theory of Machines and Mechanisms’, 3rd edition, Oxford University Press., 2003.
5. J. Hannah, and R. C. Stephens, ‘Mechanics of Machines: Elementary Theory and

Examples', 4th edition., Viva Books., 2004.

Course Outcomes: After completion of this course,
CO1: Students will recognize the concepts of kinematic links, chains, mechanisms and inversions of various mechanisms.
CO2: Students will construct the velocity and acceleration diagrams of different mechanisms.
CO3: Students will analyze power transmission devices i.e. belts, ropes and chain drives.
CO4: Students will analyse the gear profile and power transmission.

CO-PO-PSO Mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	3	2	2		1	1			3	3	
CO2	3	3	3	3	2	2		1	1			3	3	
CO3	3	3	3	2	2	2		1	1			3	3	
CO4	3	3	3	3	2	2		1	1			3	3	
AVG	3	3	3	2.75	2	2		1	1			3	3	

PI1303: Indian Art, Culture and Science/Vedic Mathematics (2-0-0)

Society State and Polity in India: State in Ancient India: Evolutionary Theory, Force Theory, Mystical Theory, Contract Theory, Stages of State Formation in Ancient India, Kingship, Council of Ministers Administration Political Ideals in Ancient India Conditions of the Welfare of Societies, The Seven Limbs of the State, Society in Ancient India, Purusartha, Varnashrama System, Ashrama or the Stages of Life, Marriage, Understanding Gender as a social category, The representation of Women in Historical traditions, Challenges faced by Women, Four-class Classification, Slavery.

Indian Literature, Culture Tradition and Practices: Evolution of script and languages in India: Harappan Script and Brahmi Script. The Vedas, the Upanishads, the Ramayana and the Mahabharata, Puranas, Buddhist And Jain Literature in Pali, Prakrit And Sanskrit, Kautilya's Arthashastra, Famous Sanskrit Authors, Telugu Literature, Kannada Literature, Malayalam Literature, Sangama Literature Northern Indian Languages & Literature, Persian And Urdu, Hindi Literature.

Indian Religion, Philosophy, and Practices: Pre-Vedic and Vedic Religion, Buddhism, Janism, Six System Indian Philosophy, Shankaracharya, Various Philosophical Doctrines, Other Heterodox Sects, Bhakti Movement, Sufi movement, Socio religious reform movement of 19th century, Modern religious practices.

Science, Management and Indian Knowledge System: Astronomy in India, Chemistry in India, Mathematics in India, Physics in India, Agriculture in India, Medicine in India, Metallurgy in India, Geography, Biology, Harappan Technologies, Water Management in India, Textile Technology in India, Writing Technology in India Pyrotechnics in India Trade in Ancient India/, India's Dominance up to Pre-colonial Times.

Cultural Heritage and Performing Arts: Indian Architect, Engineering and Architecture in Ancient India, Sculptures, Seals coins, Pottery, Puppetry, Dance, Music, Theatre drama, Painting, Martial Arts Traditions, Fairs and Festivals, Current developments in Arts and

Cultural, Indian's Cultural Contribution to the World.

II Year: II-Semester [4th Semester]

PI1401: Manufacturing Processes-II **(3-1-0)**

Introduction to Machining and Cutting Tool Geometry: Different conventions or methods of defining tool geometry; Geometry of a single point turning tool; Geometry of milling cutters and twist drill; Conversion of tool angles of a single point turning tool.

Mechanism of chip formation: Chip formation mechanism in ductile and brittle materials; Geometry and characteristics of continuous chip formation: chip reduction coefficient and cutting ratio, shear angle, cutting strain, built-up edge formation, classification of chips, shear plane and shear zone theories, orthogonal and oblique cutting, chip tool contact length.

Mechanics of chip formation: Development and action of cutting force, Analysis of cutting forces in orthogonal machining using Merchant's Circle Diagram; Working principle of measurement of cutting forces by dynamometers; Dynamometers for estimating cutting forces.

Cutting Temperature: Locations and causes of heat generation in machining; Analytical methods of evaluation of cutting temperature; Measurement of cutting temperature; Role of variation of different geometrical and process parameters on cutting temperature; Control of cutting temperature and cutting fluid application.

Tool Life: Major causes and modes of failure of cutting tools; Mechanisms of cutting tool wear; Geometry and measurement of cutting tool wear; Definition and evaluation of tool life; Taylor's tool life equation; Role of different machining parameters on tool life; Assessment and improvement of cutting tool materials.

Introduction to Machine Tools: Principle of machine tools, Brief constructional details of lathe, milling machine, drilling machine, grinding machine, Kinematic systems of lathe, milling machine.

Text and Reference Books:

1. A. B. Chattopadhyay, 'Machining and Machine Tools', Wiley Publishers.
2. A. Bhattacharyya, 'Metal Cutting – Theory and Practice', New Central Book Agency
3. M. C. Shaw, 'Metal Cutting Principles', Oxford University Press.
4. A. Ghosh, A. K. Mallik, 'Manufacturing Science', East-West Press Pvt. Ltd.
5. 'Manufacturing Processes II', Prof. A. B. Chattopadhyay, Prof. A. K. Chattopadhyay, Prof. S. Paul, Department of Mechanical Engineering, IIT Kharagpur (NPTEL Course)

Course Outcomes: After completion of this course,

CO1: Analyze the working and application of different machining processes

CO2: Estimate cutting forces, tool life, machinability for different machining applications

CO3: Ability to use the different machining and finishing processes

CO4: Ability to select suitable tools and machining operations sequence required for any given designed component

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	3	3	1	2					3	2	
CO2	3	3	3	3	3	1	1					2	2	
CO3	3	2	2	2	2	1	1					2	2	
CO4	3	3	2	3	2	2	1					3	2	
AVG	3	2.7	2.5	2.5	2.5	1	1					2.5	2	

PI1402: Engineering Economics and Accountancy (3-0-0)

Introduction to Engineering Economics: Definition and goals, Flow in an economy, Law of supply and demand, Factors influencing supply and demand, Principles of engineering economy, Engineering economy and the design process.

Time value of money: Interest, Time value equivalence, Cash flow diagrams, Present worth comparisons, Future worth comparisons, Annual worth comparisons, Rate of return analysis.

Cost Analysis: Cost concepts, Opportunity cost, Fixed cost, Variable costs, Break-even Analysis (BEA), Determination of Break-Even Point (simple problems) - Managerial Significance and limitations of BEA.

Make or Buy decisions: Introduction, Criteria for make or buy decisions, Approaches for make or buy decisions – Simple cost analysis, Economic analysis.

Analysis of alternatives: Development and classification of alternatives, IRR analysis of independent alternatives.

Evaluation of public alternatives: Benefit – Cost analysis, Quantification of project costs and benefits, Alternative selection using incremental benefit – cost analysis.

Replacement and Maintenance analysis: Types of maintenance, types of replacement problem, Determination of economic life of an asset, Replacement of an asset with a new asset – capital recovery with return and concept of challenger and defender, Simple probabilistic model for items which fail completely.

Depreciation: Methods of depreciation – Straight line method, Declining balance method, sum of the year's digits method, Sinking fund method, Service output method.

Capital: Capital and its significance, Types of Capital, Estimation of fixed and working capital requirements, Methods and sources of raising finance.

Capital Budgeting: Nature and scope of capital budgeting, features of capital budgeting proposals, Methods of Capital Budgeting: Payback method, Accounting Rate of Return, and Net Present Value Method.

Accountancy: Management Accounting - Meaning and Purpose, Financial accounting – principles, Books of accounts

Text and Reference Books:

1. Riggs, J. L., D. Bedworth, D.B, Randhawa, S.U. Engineering Economics, McGraw Hill International Edition, 1996.
2. Panneerselvam, R, "Engineering Economics", Prentice Hall of India Ltd, New Delhi,

2001.
3. Degarmo, E.P., Sullivan, W.G and Canada, J.R, “Engineering Economy”, Macmillan, New York, 2011.
4. Fundamentals of cost accounting by William N. Lanen, Shannon W. Anderson, Michael W. Maher, McGraw Hill education
5. C. S. Park, ‘Contemporary Engineering Economics’, Pearson Education, 2002.
6. N.G, Mankiw, ‘Economics: Principles and Applications’, South Western of Cengage Learing, 2007.
7. G. Donald, Newman, Jerome. P. Lavelle, ‘Engineering Economics and analysis’, Engg. Press, Texas, 2010.
8. Aryasri, ‘Managerial Economics and Financial Analysis’, TMH, 2nd edition, 2005.
9. Varshney and Maheswari, ‘Managerial Economics’, 5th edition Sultan Chand, 2003.
10. H. Craig Peterson and W. Cris Lewis, ‘Managerial Economics’, PHI, 4th Edition.
11. Domnick Salvatore, ‘Managerial Economics In a Global Economy’, Thomson, 4th Edition.

Course Outcomes: After completion of this course,

CO1: Students will understand the fundamental concepts of engineering economics and accountancy.

CO2: Students will apply the cost concepts for breakeven analysis and make or buy decisions

CO3: Students will apply the concept of time value of money to perform economic evaluation of alternatives.

CO4: Students will investigate various industrial problems through the concepts of replacement, maintenance and depreciation methods.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	2	2	2	2			2		1	3	2	1	2
CO2	3	2	2	2	3			1	1		3	3	1	2
CO3	3	2	3	2	3	1	1		1		3	3	2	3
CO4	3	2	3	3	3			1	1		3	2	2	3
AVG	**2.5**	**2**	**2.5**	**2.25**	**2.75**	**0.25**	**0.25**	**1**	**0.75**	**0.25**	**3**	**2.5**	1.5	2.5

PI1403: Production Management **(3-0-0)**

Plant and business organization: Concept, importance and principles of organization, organization chart, organization manual, Types of organization according to structure along with their advantages, disadvantages and application. Classification of business organization and ownership along with their advantages, disadvantages and application.

Production planning and control: Introduction, functions of PPC, Design of Production planning and control system.

Facility layout and location of plant: Concept and factors governing plant location, location economics, Rural vs. urban plant sites, Process layout, Product layout, Combination layout, Fixed layout, Flow pattern and material handling, Case studies on facility location and plant layout, factory simulation using open-source simulation softwares.

Capacity planning: Definition, measures of capacity, time horizon in capacity planning, capacity planning framework, alternatives for capacity augmentation, decision tree for capacity planning.

Materials management: Purchase organization, buying techniques, stores and material control, receipts and issue of materials, physical verification of stores.

Text and Reference Books:

1. M. Mahajan, 'Production planning and control', Dhanpat Rai Publishers
2. O. P. Khanna, 'Industrial Engineering and Management'.
3. S N Chary, 'Production and Operation Management', 5th Edition, Tata Mac Graw Hill.
4. J. L. Riggs, 'Production Systems: Planning, Analysis & Control', 4th Edition, John Wiley & Sons.
5. E. S. Buffa and Sarin, 'Modern Production/Operation management', 8th Edition, John Wiley & Sons.
6. Paneersalvem, 'Production & Operations Management', 2nd edition, PHI Learning Private Limited, New Delhi, 2005.

Course Outcomes: After completion of this course,

CO1: Students will identify production and service to related decisions in business organizations.

CO2: Students will demonstrate the production system using the concepts of production, planning and control.

CO3: Students will analyse production schedules and resources required for production.

CO4: Students will apply the concepts of purchase, stores and inventory management to determine material requirement decisions.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	2	2	3	1	2	1	1	3	3		3
CO2	3	2	3	2	3	2	1	1	1	2	3	3	3	3
CO3	3	2	2	2	2	2		1	1	2	3	3	3	3
CO4	3	2	2	1	2	1	1	1	1	2	2	1	3	3
AVG	**3**	**2**	**2.25**	**1.75**	**2.25**	**2**	**0.75**	**1.25**	**1**	**1.75**	**2.75**	**2.5**	**2.25**	**3**

ME1409: Design of Machine Elements **(3-0-0)**

Introduction: Design procedure, Selection of preferred sizes, Aesthetic and ergonomic considerations in design, Manufacturing considerations in machine design, Engineering materials and their properties, Effect of alloying elements and heat treatment on properties of steels, Materials selection in machine design. IS coding of steel and cast iron, Simple stresses in machine parts, Torsional and bending stresses, Dynamic loads.

Design against fluctuating load: Stress concentration, Endurance limit, Factors affecting endurance limit, Fatigue failure, S-N diagram, Design for reversed stresses and cumulative damage, Fluctuating stresses, Soderberg, Gerber, Goodman and Modified Goodman criteria, Combined stresses.

Design of riveted joints, welded joints, bolted joints, cotter joint, knuckle joint. Design of pressure vessels and pipe joints. Design of keys, couplings, shafts, levers, columns, studs and power screw. Design of belt and chain drives, pulleys, springs, clutches and brakes.

Text and Reference Books:

1. V.B. Bhandari, ‘Design of Machine Elements’, Tata McGraw Hill Co.
2. Sharma and Agrawal, ‘Machine Design’, S.K. Kataria & Sons.
3. U C Jindal, ‘Machine Design’, Pearson Education.
4. Sharma and Purohit, ‘Design of Machine Elements’, PHI Learning Private Limited, New Delhi,
5. M.F. Spott, ‘Design of Machine Elements’, Pearson Education.
6. Maleev and Hartman, ‘Machine Design’, CBS Publishers.
7. Joseph E. Shigely, ‘Mechanical Engineering Design’, McGraw Hill Education.
8. Juvinal and Marshek, ‘Elements of Machine Component Design’, John Wiley & Sons.

Course Outcomes: After completion of this course,

CO1: Students will demonstrate the concepts of simple stresses, strains and elastic constants.

CO2: Students will design the machine elements against static and fluctuating loads.

CO3: Students will design the cotter joint, knuckle joint, rivet joints.

CO4: Students will design the Pressure vessels, keys, couplings, shafts, columns, screw jack, belt and chain drives, pulleys, springs, clutch and brakes etc.

CO-PO-PSO Mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	3	2	2		1	1			3	3	
CO2	3	3	3	3	2	2		1	1			3	3	
CO3	3	3	3	2	2	2		1	1			3	3	
CO4	3	3	3	3	2	2		1	1			3	3	
AVG	3	3	3	2.75	2	2		1	1			3	3	

ME1410: Fluid Mechanics and Machinery (3-0-0)

Introduction: Concept of continuum; Fluid properties: Viscosity, compressibility, surface tension and capillarity; Types of fluid: real and ideal, Newtonian and Non-Newtonian compressible and incompressible.

Fluid Statics: Fluid pressure and its measurement; manometers: simple and differential types, pressure gauges; hydrostatic force and centre of pressure on plane submerged surfaces. Buoyancy and Floatation: buoyant force and centre of buoyancy, meta-centre and meta-centric height, determination of meta-centric height, conditions of equilibrium of floating and submerged bodies.

Fluid Dynamics: Forces influencing motion of fluid, Euler's equation of motion, Bernaulli's equation, momentum and moment of momentum equations, kinetic energy and momentum correction factors, introduction to Navier-Stroke's Equation.

Introduction: Classification of Fluid Machineries, Dynamic Action of Fluid Jet: Impact of fluid jet on fixed and moving flat plates, Impact of jet on fixed and moving curved vanes, flow over radial vanes, Jet Propulsion, Euler's fundamental equation, Degree of reaction.

Hydraulic Turbines: Introduction, classification, Impulse Turbine: constructional details, velocity triangles, power and efficiency calculations. Reaction Turbines: constructional details, working principle, velocity triangles, power and efficiency calculations, draft tube, cavitation, governing of turbines, performance characteristic curves.

Positive Displacement Pumps: Reciprocating Pump: working principle, classification, slip, indicator diagram, effect of friction and acceleration, theory of air vessel, performance characteristic curves; gear oil pumps and screw pump.

Roto-dynamic Pumps: Introduction, classification, centrifugal pump: main components, working principle, velocity triangles, effect of shape of blade, specific speed, heads, power and efficiency calculations, minimum starting speed, multistage pumps, performance characteristics, comparison with reciprocating pump.

Text and Reference Books:

1. Fluid Mechanics by Frank M. White, McGraw Hill Education
2. Fluid Mechanics and Hydraulic Machines by R. K. Bansal, Laxmi Publication
3. Engineering Fluid Mechanics by K.L Kumar, S. Chand Publication

Course Outcomes: After completion of this course,

CO1: Students will demonstrate the concepts of fluid mechanics and fluid statics.
CO2: Students will describe the concepts of fluid dynamics.
CO3: Students will analyse various hydraulic turbines.
CO4: Students will analyse various hydraulic pumps.

CO-PO-PSO Mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	3		2	1					3	3	
CO2	3	3	3	3		2	1					3	3	
CO3	3	3	3	2		2	3					3	3	
CO4	3	3	3	3		2	3					3	3	
AVG	3	3	3	2.75		2	2					3	3	

III Year: I-Semester [5th Semester]

PI1501: Computer Aided Design and Manufacturing (CAD/CAM) (3-0-0)

Introduction: CAD/CAM Defined, Product Cycle and CAD/CAM, Automation and CAD/CAM, Computer Technology.

Basic Concepts: Design Process, Application of Computer for Design, Hardware and Software requirement of CAD, Benefits of CAD, Engineering Application of CAD, Hardware in CAD.

Finite Element Method: Introduction to Finite Elements Method, General descriptions, Concept of FEM - discretization and interpolation function, steps on finite element analysis procedure, Introduction to Auto CAD and modeling by using AutoCad.

Conventional Numerical Control: Introduction, Basic components of NC system, NC procedure, NC Coordinate System, NC Motion Control Systems, Punched Tape, Tape Coding and Format.

NC Part Programming & Computer Control in NC: Manual Part Programming, Computer assisted Part Programming, APT Language, Computer Control System, Direct Numerical Control. CNC coding for various machining operations.

Flexible Manufacturing Systems: Introduction, Different FMS Levels, FMS implementation, Automated Guided Vehicle Systems, AGVS Controller Structure, AGVS Classification, Applications

Text & Reference Books:

1. M. Groover and E. Zimmers, 'CAD/CAM', Pearson.
2. P. N. Rao, 'CAD/CAM – Principles and Applications', McGraw Hill Inc., New Delhi.
3. Ibrahim Zeid, 'CAD/CAM Theory and Practice', McGraw Hill Inc., New Delhi, 2014.
4. P.Radhakrishnan and S.Subramanyan, 'CAD/CAM/CIM', New Age International, 2016.

Course Outcomes: After completion of this course,

CO1: Students will recognize the basic foundation in computer aided design and the fundamentals used to create and manipulate geometric models

CO2: Students will describe the basic foundation in computer aided manufacturing and learning the working principles of machines, coding system and part programming and FMS

CO3: Students will apply the concept of CAD in creating geometric modeling by using modern tools and applying the CAM concept in CNC machines

CO4: Students will analyze the requirement of CAD/CAM and possible solutions for a production shop for a manufacturing product

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	1	2							2	1	1
CO2	2	3	2	1	3							2	2	2
CO3	3	3	3	3	3							2	3	1
CO4	3	3	3	3	2							2	2	2
AVG	2.75	3.0	2.5	2.0	2.5							2	2	1.5

PI1502: Supply Chain Management **(3-0-0)**

Introduction and overview of supply chain management, Inbound and outbound logistics, Supply chain as a source of competitive advantage. Supply chain drivers and matrices. Case studies

Network design supply chain: Role of network design in supply chain, Factors influencing the supply chain network design, warehouse location problem, capacitated warehouse location problem, Gravity location model, plant location model, Case studies.

Network design for global supply chain: Offshoring, Global supply chain risks, network design through decision tree, decision for offshoring or onshoring, Case studies.

Demand forecasting and aggregate planning: Time series forecasting, forecast error estimation, Aggregate planning using excel spreadsheet.

Inventory management in supply chain: various costs in inventory management, economic order quantity, production lot size, lot size with multiple products, quantity discount models, multi echelon inventory models, inventory models encountering the uncertainty in supply chain, Sourcing decisions in supply chain, Pricing Strategies.

Technology in SCM: Information Technology infrastructure, Web services and business processes, Blockchain, Internet of Things (IoT)

Text and Reference Books:

1. J. R. Tony Arnold, Stephen N. Chapman, Lloyd M. Clive, 'Introduction to Materials Management', 7th Edition
2. F. Robert Jacobs, William Berry, D. Clay Whybark, Thomas Vollmann, 'Manufacturing Planning and Control for Supply Chain Management',
3. David Simchi-Levi, Philip Kaminsky, Edith Simchi-Levi, 'Designing and Managing the Supply Chain 3e with Student CD',
4. Sunil Chopra, Dharam Vir Kalra, 'Supply Chain Management: Strategy, Planning, & Operation', 7th Edition, Pearson.

Course Outcomes: After completion of this course,

CO1: Students will define the supply chain management, structure, entities, issues and drivers.

CO2: Students will design the local and global supply chain networks using softwares.

CO3: Students will choose appropriate inventory management policy using demand

estimates.
CO4: Students will demonstrate applications of IT tools in supply chain management.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	2	2	3	1	2	1	1	3	3		3
CO2	3	2	3	2	3	2	1	1	1		3	3	3	3
CO3	3	2	2	2	2	2		1	1		3	3	3	3
CO4	3	2	2	1	2	1	1	1	1		2	1	3	3
AVG	3	2	2.25	1.75	2.25	2	0.75	1.25	1	0.25	2.75	2.5	2.25	3

PI1503: Product Design and Development (3-0-0)

Introduction: Definition of product, product design and product development, Significance of product design, Characteristics of successful product development, functions of a firm, Challenges of product development.

Development Processes and Organizations: Generic development process, phases of generic development process, activities of concept development process, product development organizations.

Product Planning, Product Specifications and Product Architecture:

Product planning process, Identifying Customer needs: Steps of Identifying Customer needs. Product Specifications: Steps of establishing target specifications, Steps of setting final specifications, Product architecture: Definition of Product architecture, types of modularity, steps of establishing the architecture.

Concept Generation, Concept Selection and Concept Testing: Concept generation: Steps of concept generation method, Concept selection: Steps of concept selection, Concept testing: steps of concept testing.

Industrial Design, Design for Manufacturing, Prototyping and Robust Design:

Industrial design: Definition of Industrial design, Phases of industrial design process, Design for Manufacturing (DFM): Definition of DFM, Steps of DFM process, Prototyping: Definition of prototype, Types of prototype, Process with prototyping, Steps Involved in planning for prototypes, Robust Design: Definition of robust design, steps of robust design process.

Patents and Intellectual Property, Product Development Economics and Managing Projects:

Patents & Intellectual Property:Types of intellectual property, process for pursuing a patent, Product Development Economics: Elements of economic analysis, Steps involved in economic analysis process, Managing Projects: Definition of project management, representing tasks.

Text and Reference Books:

1. Karl. T. Ulrich and Steven D. Eppinger, ‘Product Design & Development’, TMH, Delhi.

2. Kevin Otto and Kristin Wood, ‘Product Design’, Pearson Education.
3. Chitale & Gupta, ‘Product Development’, Tata McGraw Hill.
4. Chitale & Gupta, ‘Product Design and Manufacturing’, PHI, Delhi.
5. Prashant Kumar, ‘Product Design: Creativity, Concepts and Usability’, PHI, Delhi.
6. Imad Moustapha, ‘Concurrent Engineering in Product Design and Development’, New Age.
7. J. G. Monks, ‘Operations Management’, McGraw Hill.
8. Ulrich & Eppinger, ‘Product Design and Development’, TMH Delhi.
9. R. L. Francisand and J. A. White, ‘Facility Layout and Location’, Prentice Hall of India

Course Outcomes: After completion of this course,

CO1: Students will come to know about the basic concepts of product design, development process and its architecture.

CO2: Students will able to understand the importance of product design process in the development of various industrial components

CO3: Students will understand the various aspects of design such as industrial design, design for manufacturing and robust design.

CO4: Students will construct the industrial design and demonstrate the basics of intellectual property.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	1	2		2		2		3		3
CO2	3	2	2	2	3	2	2	2	3	2	2	3		3
CO3	3	2	3	3	3	2		1	3	2	2	2		3
CO4	3	3	3	1	1	2		3	3	2	2	2		3
AVG	3	2.25	2.25	1.75	2	2	0.5	2	2.25	2	1.5	2.5		3

PI1504: Operations Research (3-1-0)

Scope and application of operation research. Linear programming, graphical and simplex method. Special cases in linear programming problems. Sensitivity analysis- graphical and Algebraic. Duality in linear programming.

Specially structured linear programming problems: Transportation - North west corner method, Least cost method, Penalty method. Methods for optimality check - Modified distribution (MODI) method, Vogel’s approximation method (VAM), Assignment models - Hungarian method, Dynamic programming, Replacement problems,

Queuing theory (single and double channel). Sequencing model (n jobs – 2 machines, n jobs - 3 machines): Johnson’s Algorithm, CPM and PERT and CPM- crashing networks.

Inventory models with probabilistic demands and area, quantity constraints, Game theory (competitive strategies).

Text and Reference Books:

1. D S Hira, Prem Kumar Gupta, ‘Problem in Operation Research- Principles & Solution’,

2. K. Swarup, P. K. Gupta, M. Mohan, ‘Operations Research’,
3. N. D. Vohra, ‘Quantitative Techniques in Management’,
4. W. L. Winston, ‘Operations Research Applications and Algorithms’,
5. H. A. Taha, ‘Operations Research An Introduction’,
6. F. S. Hillier, G. J. Lieberman, ‘Introduction to operations research’,

Course Outcomes: After completion of this course,
CO1: Students will formulate linear programming problems for industrial cases and propose optimum solutions.
CO2: Students will calculate the optimal solution for transportation and assignment problems.
CO3: Students will analyse queuing models and project management problems.
CO4: Students will analyse various inventory models.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	2	2	3	1	2	1	1	3	3		3
CO2	3	2	3	2	3	2	1	1	1		3	3	3	3
CO3	3	2	2	2	2	2		1	1		3	3	3	3
CO4	3	2	2	1	2	1	1	1	1		2	1	3	3
AVG	**3**	**2**	**2.25**	**1.75**	**2.25**	**2**	**0.75**	**1.25**	**1**	**0.25**	**2.75**	**2.5**	**2.25**	**3**

HSIR13: Fundamentals of Intellectual Property Right (3-0-0)

Concept and Relevance of IPR: Concept of Intellectual Property; Concept of IPR; Need, Relevance of IPR; Role of IPR in the Socio-Economic Development of country, Contribution of IPR in the technological innovation and growth; Implication of IPR for in globalized scenario

Patent: Patent: Meaning of Patent, Concept of Novelty, Inventiveness and Utility; Patentable Subject matter, Patentability criteria, Duration of patent, Inventions not patentable; Process and Product Patents.

Copyright: Meaning & Scope, Nature of copyright, copyright works, Author & Ownership of Copyright, duration. Trademark: Definition of Trademark; Kinds of marks - brand names, logos, signatures, symbols, well known marks, Domain Names.

Other forms of Intellectual Property: Industrial Design: Its novelty and originality, Registrable design, Items not protected under Design. Layout - Semiconductor Integrated Circuits, Design of Plant varieties, Geographical Indications, Undisclosed information.

Emerging Dimensions of IPR: WTO, TRIPS & International IP Regime, Future Challenges and Issues relating to globalization Global Trade and IPR, Issues in Commercialization of IPR, Recent government policies and interventions relating to IPR 2 Course Outcome Knowledge of various laws and issues relating to the various aspects of IPR would enable the students to apply the same in real life situations

Text and Reference Books:

1. Prabuddha Ganguli, (2001): Intellectual Property Rights. Tata McGraw Hill
2. W.R. Cornish, (2013): Intellectual Property: Patents, Copyright, Trade Marks and Allied Rights. Sweet and Max Well London
3. Blakeney, (1996), Trade-Related Aspects of Intellectual Property Rights: A Concise Guide to the TRIPS Agreement. Sweet and Max Well London
4. Satyawrat Ponkshe, (1991) The management of intellectual property: Patents, designs, trade marks & copyright, Exclusive distributors, UBSPD
5. Ahuja V K, Law Relating to Intellectual Property Rights, LexisNexis, 2017
6. Gopalakrishnan and Agitha, Principles of Intellectual Property, Eastern Book Co., 2006

Course Outcomes: On successful completion of the course,

CO1: Students will illustrate the concept of Intellectual Property and patents.

CO2: Students will analyze copyrights and responsibilities of the holder of Patent, Copyright, Trademark, Industrial Design, etc.

CO3: Students will distinguish various forms of IPRs.

CO4: Students will identify the procedure to protect different forms of IPRs at national and international levels.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1						1	1	2	1	1	3	3	2	2
CO2						1	1	1	1	1	3	3	2	2
CO3						1	1	1	1	1	3	3	2	2
CO4						1	1	1	1	1	2	1	2	2
AVG						**1**	**1**	**1.25**	**1**	**1**	**2.75**	**2.5**	**2**	**2**

Professional Elective – I **(3-0-0)**

PI1505: Surface Engineering

Introduction to Surface Engineering: Definition and scope, Surface dependent properties and failures of engineering components, Structure and types of interfaces, Surface energy and related equations

Surface Modification by LASER: Fundamentals of Light Amplification by Stimulated Emission of Radiation (LASER), LASER assisted microstructural modification: transformation hardening, melting and resolidification, alloying and cladding

Surface Coatings: Deposition of coatings by flame spray, plasma spray, high velocity oxy-fuel (HVOF) spray, cold spray, deposition of coatings by physical vapour deposition (PVD) techniques like thermal/electron beam evaporation; sputtering (direct current (DC), radio frequency (RF), magnetron), deposition of coatings by chemical vapor deposition (CVD) and plasma enhanced chemical vapour deposition (PECVD); plasma and ion beam assisted surface modification

Characterization of Coatings: Assessment of surface morphology by optical and scanning electron microscopy, chemical composition by energy dispersive spectroscopy and structure of coatings by X-ray diffraction, assessment of coating hardness by Vicker's and nanoindentation tests, assessment of friction and wear of coating by pin-on-disc tests, assessment of surface roughness by surface profilometry and thickness of coating by ball cratering and fractography tests, assessment of adhesion of coating by scratch and indentation adhesion tests

Performance Evaluation of Coated Product: Performance evaluation of PVD TiN coated and CVD diamond coated cutting tools.

Text and Reference Books:

1. K.G. Budinski, ‘Surface Engineering for Wear Resistances’, Prentice Hall, Englewood Cliffs, 1988.
2. Rointan F. Bunshah, William Andrew, ‘Handbook of Hard Coatings - Deposition Technologies, Properties and Applications’, 1st Edition.
3. M. Ohring, ‘The Materials Science of Thin Films’, Academic Press Inc, 2005.
4. Surface Engineering and Coating Technology by Prof. I. Manna, Department of Metallurgy and Materials Engineering, IIT Kharagpur and Prof. S. K. Ray, Department of Physics & Meteorology, IIT Kharagpur (NPTEL Course)
5. Technology of Surface Coating by Prof. A. K. Chattopadhyay, Department of Mechanical Engineering, IIT Kharagpur (NPTEL Course)

Course Outcomes: After completion of this course,

CO1: Students will be able to understand different surface engineering techniques and their importance and applications.

CO2: Students will be able to learn different laser-based surface modification techniques and applications

CO3: Students will be able to learn different surface coating technologies

CO4: Students will be able to evaluate performance of surface coated components

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	2	1	1					3	3	
CO2	3	3	3	3	2	1	1					3	3	
CO3	3	2	3	3	2	1	1					2	2	
CO4	3	3	3	3	2	1	1					3	3	
AVG	3	2.7	3	2.7	2	1.0	1.0					2.7	2.7	

PI1506: Precision Engineering

Introduction to precision engineering: Concept of precision, accuracy, and smoothness, Need for having a high precision, Four classes of achievable machining accuracy, Precision machining, High-precision, Ultra-precision processes, Application of precision machining.

Tool materials and tool development for precision machining: Coated and laminated carbides, Ceramics, Diamond, Cubic Boron Nitride, Cutting insert fabrication: powder metallurgy, surface texturing, surface coating, cutting tool geometry and its importance.

Ultra-precision machine elements: Guide-ways, Drive-system, Friction drives, Linear motor drive, Spindle drive, Hydrodynamic and hydrostatic bearings: Principle of rolling element bearings, Bearing life, Construction of lubricated sliding bearings, Principle of hydrodynamic bearings, Hydrodynamic thrust bearings. Design of hydrostatic bearings, Hybrid fluid bearings, Aerostatic bearings, Hybrid gas bearings.

Error during precision machining: Error due to compliance of machine-fixture-tool-workpiece, Theory of location, Location error, Clamping and setting error, Error due to geometric inaccuracy of machine tools, Errors due to tool wear, Error due to thermal effects.

Micro-electro-mechanical systems: Characteristics and principles, Materials and design, Application of MEMS, Fabrication and Micro-manufacturing processes, Clean rooms, Design and construction of clean rooms.

Text and Reference Books:

1. V.C. Venkatesh and I. Sudin, 'Precision Engineering', Tata McGraw Hill Co., 2007.
2. R. L. Murthy, 'Precision Engineering', New Age International, 2009.
3. S. Kalpakjian, 'Manufacturing Engineering and Technology' , 3rd Edition, Addison-Wesley Publishing Co.
4. H. Nakzavawa, 'Principles of Precision Engineering', Oxford University Press, 1994.

Course Outcomes: After completion of this course,
CO1: Students will be able to know the concepts of precision engineering and its importance.
CO2: The requirements of machine network elements to achieve precision in the components.
CO3: The principles of various precision engineering processes and apply them in actual fields.
CO4: Evaluate the part and machine tool accuracies.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	3	1		2				3	3	
CO2	3	3	3	2	2	1		2				3	3	
CO3	3	2	2	1	3	1		2				3	3	
CO4	3	3	3	3	3	1		3				3	3	
AVG	3	2.5	2.5	1.75	2.75	1		2.25				3	3	

PI1507: Strategic Management

Introduction: meaning nature, scope, and importance of strategy; Model of strategic management, Strategic Decision-Making Process. Corporate Governance: Composition of the board, Role and Responsibilities of the board of directors, Trends in corporate governance, Corporate Social Responsibility. Case Studies and Latest Updates.

Environmental Scanning: Understanding the Macro Environment: PESTEL Analysis, Industrial Organization (IO) & the Structure Conduct Performance (SCP) approach, Porter's Five Forces Model, Understanding the Micro Environment: Resource Based View (RBV) Analysis, VRIO Framework, Using resources to gain Competitive advantage & its sustainability, Value Chain Analysis. Case Studies and Latest Updates.

Strategy Formulation: Situational Analysis using SWOT approach Business Strategies: Competitive Strategy: - Cost Leadership, Differentiation & Focus, Cooperative Strategy: - Collusion & Strategic Alliances Corporate Strategies: Directional Strategy: Growth strategies, Stability Strategies & Retrenchment Strategies. Corporate Parenting Functional Strategies: Marketing, Financial, R&D, Operations, Purchasing, Logistics, HRM & IT. The sourcing decision: Outsourcing & offshoring Case Studies and Latest Updates.

Strategy Choice and Analysis: Scenario Analysis Process, Tools & Techniques of strategic Analysis: BCG Matrix, Ansoff Grid, GE Nine Cell Planning Grid, McKinsey’s 7’S framework. Case Studies and Latest Updates. Strategy implementation: Developing Programs, Budget and Procedures, Stages of Corporate Development, Organizational Life cycle, Organizational Structures: Matrix, Network & Modular/Cellular; Reengineering and Strategy implementation, Leadership and corporate culture, Case Studies and Latest Updates.

Strategy Evaluation & Control: Evaluation & Control process, Measuring performance: types of controls, activity based costing, enterprise risk management, primary measures of corporate performance, balance scorecard approach to measure key Performance, responsibility centers, Benchmarking, Problems in measuring Performance & Guidelines for proper control. Strategic Audit of a Corporation. Case Studies and Latest Updates.

Text and Reference Books

1. Wheelen, L. Thomas and Hunger, David J., ‘Concepts in Strategic Management and Business Policy’, Pearson Education.
2. Stewart Clegg, Chris Carter, Martin Kornberger & Jochen Schweitzer, ‘Strategy - Theory and Practice’, SAGE Publishing India.
3. Kazmi, Azhar, ‘Business Policy and Strategic Management’, McGraw-Hill Education.
4. David, Fred, ‘Strategic Management: Concepts and Cases’, PHI Learning.
5. Thomson, Arthur A. and Strickland, A. J., ‘Strategic Management: Concept and Cases’, McGraw Hill Education.
6. Jauch, L.F., and Glueck, W.F., ‘Business Policy and Strategic Management’, McGraw-Hill Education.

Course Outcomes: After completion of this course,

CO1: Students will formulate organizational vision, mission, goals, and values.

CO2. Students will develop strategies and action plans to achieve an organization's vision, mission, and goals.

CO3. Students will develop powers of managerial judgment, how to assess business risk, and improve ability to make sound decisions and achieve effective outcomes.

CO4. Students will consider the ethical dimensions of the strategic management process.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	2	1	1	3	1	2	1	1	1	3		3
CO2	2	2	3	2	2	2	1	2	1	1	3	3		3
CO3	2	3	3	2	2	2	3	2	1	1	3	3		3

CO4	2	2	3	1	1	3	2	3	1	1	2	1		3
AVG	10	2.25	2.75	1.5	1.5	2.5	1	1.25	1	1	2.75	2.5		3

PI1508: Digital Manufacturing

Overview of digital manufacturing processes, What makes a manufacturing process “digital”, The 10 disruptive principles of digital manufacturing processes, Industry 4.0, smart manufacturing, artificial intelligence, machine learning, deep learning, internet of things, machine vision, virtual reality, augmented reality.

Throughput analysis, Shop layout optimization, Line balancing and simulation of automated machinery and production lines, Work instructions and automated process sheet, Workplace safety analysis

Additive Manufacturing processes – Engineering polymers, metals, ceramics: Stereolithography, Selective Laser Sintering, Fused Deposition Modelling, Polyjet, LENS, Layered object manufacturing, Additive Manufacturing processes for advanced materials: Electronic Materials, Bioprinting, Food Printing, Material properties, Programmable Assembly, Digital Assembly, Digital Bending

Fundamentals of geometric representations for digital manufacturing: Solid representations, Boundary representations, Function representations, Voxel representations

Algorithmic design for digital manufacturing: Parametric Models, Vibrational Geometry, Generative models, Topology optimization, Industrial Safety, Liability and intellectual property, Environmental impact,

Internet of Things: Advantages and Challenges.

Text and Reference Books:

1. Chua Chee Kai, Leong Kah Fai, ‘Rapid Prototyping: Principles & Applications’, World Scientific, 2003.
2. Ian Gibson, David W Rosen, Brent Stucker., ‘Additive Manufacturing Technologies: Rapid Prototyping to Direct Digital Manufacturing’, Springer.
3. Groover, M.P. and Zimmers, E.W. Jr., ‘CAD/CAM: Computer-aided Design and Manufacturing’, Prentice-Hall of India Private Ltd., New Delhi.

Course Outcomes: After completion of this course,

CO1: To understand the need of digital manufacturing over conventional manufacturing processes

CO2: To understand the process chain of digital manufacturing processes

CO3: To analyze the parametric effects in various digital manufacturing processes

CO4: To be able to design for digital manufacturing of different engineering applications

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	3	2	2	1	1	1				3	3	1
CO2	3	3	3	3	3	1	1	1				3	3	1
CO3	3	3	2	3	3	1	1	1				3	2	1

CO4	3	3	2	3	2	1	1	1				2	2	1
AVG	**3**	**2.7**	**2.5**	**2.7**	**2.5**	**1.0**	**1.0**	**1**				**2.7**	**2.5**	**1**

III Year: II-Semester [6th semester]

PI1601: Non Traditional Manufacturing Processes (3-0-0)

Needs of manufacturing industries and the concept of surface integrity. The role of newer & innovative processes for the solutions

Impact erosion and the evaluation for impregnation of foreign bodies. Theory and application of Abrasive Jet, Water Jet, Abrasive Flow, Ultrasonic, Total Form Machining and Low stress Grinding.

Theory and applications of Chemical Processing: Chemical Machining, etching of semiconductors, coating and Electroless forming and CVD.

Theory and applications of Electrochemical machining and grinding including, Electrochemical sharpening, polishing, honing and deburring, surface treatment, coating and Electroforming.

Thermal energy methods of material processing (machining, welding, cladding, alloying and heat-treatment): Electro-discharge, Laser, Electron beam, Plasma arc, Plasma spray, Ion beam and PVD.

Hybridization of processes for improving process capability and surface integrity. Explanation of some processes like ECDG, ultrasonic and Laser assisted processing, LIGA & SLIGA etc.; Generic manufacturing and introduction to rapid prototyping.

P-component: Experiments on EDM, ECM, laser, etc.

Text and Reference Books:

1. Pandey and Shan, ‘Modern Machining Processes’, Tata McGraw-Hill Education.
2. P.K. Mishra, ‘Non-conventional Machining’, Narosa Publishing House Publisher.
3. Ghosh and Mallik, ‘Manufacturing Science’, Prentice Hall PTR.
4. Gary F. Benedict, ‘Non-traditional manufacturing processes’, CRC Press.
5. T. Jagadeesha, ‘Non-Traditional Machining Processes’, I K International Publishing.

Course Outcomes: After completion of this course,

CO1: Students will identify various nontraditional manufacturing methods which are applicable for difficult-to-cut materials, defense, aerospace and other engineering sectors.

CO2: Students will interpret the process parameters and the applicability of different materials that are suitable for various non-traditional methods.

CO3: Students will evaluate the material removal rate, surface finish achievable by various non-traditional manufacturing processes in relation to the process parameters.

CO4: Students will use advance coating technology in various fields of industrial applications.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	2	2	3	3	3		1	2		3	3	2
CO2	2	3	2	3	3	2	3		1	2		3	3	2
CO3	2	3	2	3	3	1	1		1	2		3	3	2
CO4	2	2	2	2	3	2	3		1	2		3	3	2

AVG	2	2.5	2	2.5	3	2	2.5		1	2		3	3	2

PI1602: Metal Forming **(3-0-0)**

Introduction to metal forming, classification of metal forming processes, temperature, strain rate, friction and lubrication effects in metal forming processes

Elastic and plastic deformation, mechanism of plastic deformations, yield criteria, hardening law, flow rule, Anisotropy, Anisotropic yield criterion, Introduction to crystal plasticity

Bulk metal forming processes: Rolling, Forging, Extrusion, rod or wire drawing etc., Mechanics of Forming, load estimation and defects etc and Workability.

Sheet metal forming processes, Bending, stamping, Deep drawing, stretch forming, spinning and incremental forming etc., load estimation and defects analysis, formability and forming limit diagram.

Slab, upper and lower bound techniques, slip line field theory, mechanics of large deformation and nonlinear continuum mechanics, Finite element method for analyses of bulk and sheet metal forming processes.

Miscellaneous and Non-conventional metal forming processes, High energy rate forming processes, explosive, electro-hydraulic and electro-magnetic forming etc., advanced metal forming processes

Text and Reference Books:

1. G. W. Rowe, ‘Principles of industrial metalworking processes’, CBS Publishers & Distributors, 2005.
2. A. Ghosh and A. K. Mallik, ‘Manufacturing Science’, 2nd Edition, East West Press Pvt. Ltd., 2010.
3. B. Aviitzur, ‘Metal forming processes and analysis’, McGraw Hill, 1968.
4. G. E. Dieter, ‘Mechanical Metallurgy’, Tata McGrawhill, 1998.
5. Hosford W.F and Caddell, R.M, ‘Metal Forming Mechanics and Metallurgy’, Prentice Hall, 1983.
6. D.W.A. Rees, ‘Basic Engineering plasticity’, Elsevier, 2006.

Course Outcomes: After completion of this course,

CO1: Understand theoretical concepts of metal forming and plastics processing

CO2: Apply knowledge to evaluate the behavior of sheet metal during the Forming processes

CO3: Analyze the force and power requirement in the different types of metal forming processes

CO4: Ability to design process and machine for metal forming of any given new engineering component

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	2	1	1					3	3	1
CO2	3	3	3	3	3	1	1					3	2	
CO3	3	2	3	3	3	1	1					2	2	1

CO4	3	3	3	3	2	1	1					2	3	
AVG	3	2.7	3	2.7	2.5	1.0	1.0					2.5	2.5	0.25

PI1603: Design of Production Tooling (3-1-0)

Work holding device: Purpose and function of work holder, Principles of jig and fixture design, Method of location, 3-2-1 Method of location, Principles of pin locations, Locating devices, Type of clamping devices

Jig bushes, Types of jigs, Jig design Problems

Classification of fixtures, Milling fixtures, Turning fixtures, Boring fixtures, Fixture design Problems

Press work die design: Classification of presses, Classification of dies, Centre of pressure, Cutting action in die, Die clearances, Cutting forces in die, Stock stop pilots

Piercing die design, Blanking die design, Compound die design, Scrap-strip layout for blanking, Evolution of progressive die.

Types of bending dies, Design problems, Deep drawing, Deep drawing die design problems.

Text and Reference Books

1. P. H. Joshi, ‘Jigs and Fixtures, 2nd Edition, Tata McGraw Hill, New Delhi, 2004.
2. Donaldson, Lecain and Goold, ‘Tool Design’, 3rd Edition, Tata McGraw Hill, 2000
3. K. Venkataraman, ‘Design of Jigs Fixtures & Press Tools’, Tata McGraw Hill, New Delhi, 2005.
4. Kempster, ‘Jigs and Fixture Design’, 3rd Edition, Hoddes and Stoughton, 1974.

Course Outcomes: After completion of this course,

CO1: Students will design the jigs and fixtures.

CO2: Students will design and analyse the blanking die.

CO3: Students will design bending dies.

CO4: Students will design deep drawing dies.

CO-PO-PSO Mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	3	2	2		1	1			3	3	
CO2	3	3	3	3	2	2		1	1			3	3	
CO3	3	3	3	2	2	2		1	1			3	3	
CO4	3	3	3	3	2	2		1	1			3	3	
AVG	3	3	3	2.75	2	2		1	1			3	3	

Professional Elective II (Sixth semester) (3-0-0)

PI1604: Energy Conservation and Management

Energy Scenario: Commercial and Non-commercial energy, primary energy resources, commercial energy production, final energy consumption, Indian energy scenario, Sectoral energy consumption (domestic, industrial and other sectors), energy needs of growing economy, energy intensity, long term energy scenario, energy pricing, Energy security, energy conservation and its importance, energy strategy for the future, Energy Conservation Act 2001 and its features.

Basics of Energy its various forms and conservation: Electricity basics – Direct Current and Alternating Currents, electricity tariff, Thermal Basics-fuels, thermal energy contents of fuel, temperature and pressure, heat capacity, sensible and latent heat, evaporation, condensation, steam, moist air and humidity and heat transfer.

Evaluation of thermal performance – calculation of heat loss – heat gain, estimation of annual heating & cooling loads, factors that influence thermal performance, analysis of existing buildings setting up an energy management Program and use management – electricity saving techniques

Energy Management & Audit: Definition, energy audit, need, types of energy audit. Energy management (audit) approach-understanding energy costs, Bench marking, energy performance, matching energy use to requirement, maximizing system efficiencies, optimizing the input energy requirements, fuel and energy substitution, energy audit instruments and metering

Financial Management :Investment-need, appraisal and criteria, financial analysis techniques simple payback period, return on investment, net present value, internal rate of return, cash flows, risk and sensitivity analysis; financing options, energy performance contracts and role of Energy Service Companies (ESCOs).

Text and Reference Books:

1. W. F. Kenny, ‘Energy Conservation in Process Industry’,
2. Amlan Chakrabarti, ‘Energy Engineering and Management’, Prentice hall India. 2011.
3. C. B. Smith, ‘Energy Management Principles’, Pergamon Press, New York.
4. W. C. Turner, ‘Energy Management Hand Book’, John Wiley and sons.

Course Outcomes: After completion of this course,

CO1: To understand the need of energy conversion and management

CO2: To understand the basics of energy its various forms and conservation

CO3: To be able to evaluate the thermal performance

CO4: To be able to do energy management & audit, financial management

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	3	2	2	1	1	1				3	3	1
CO2	3	3	3	3	3	1	1	1				3	3	1
CO3	3	3	2	3	3	1	1	1				3	2	1
CO4	3	3	2	3	2	1	1	1				2	2	1
AVG	3	2.7	2.5	2.7	2.5	1.0	1.0	1				2.7	2.5	1

PI1605: Concurrent Engineering

Introduction: Product development objectives and product development cycle, background and challenges faced by modern production environment, sequential engineering process. Concurrent Engineering: Definition need and utility, objectives of CE, benefits of CE, life cycle design of products, life cycle costs, support for CE: classes of support for CE activity, CE organizational, structure CE, team composition and duties, computer based support, CE implementation process. Collaborative product development.

Design Product for Customer: Industrial design, quality function deployment, translation process of quality function deployment (QFD). Modeling of concurrent engineering design: compatibility approach, compatibility index, implementation of the compatibility model, integrating the compatibility concerns.

Product life cycle: Lifecycle design of products, Implementation of PLM, Responsibility for PLM, Components of PLM, Life cycle problems to resolve, Components of PLM, Product lifecycle activities, Product organizational structure, Human resources in product lifecycle, System components in lifecycle.

Creative design methods, product family themes, design axioms, robust design: Taguchi design methods

Quality by Design: Quality engineering & methodology for robust product design, parameter and tolerance design, quality loss function and signal to noise ratio for designing the quality, experimental approach, design for reliability, life cycle serviceability design, design for maintainability, design for economics, decomposition in concurrent design, concurrent design case studies.

Text and Reference Books:

1. Menon, ‘Concurrent Engineering’, Chapman & Hall.
2. Prasad, ‘Concurrent Engineering Fundamentals: Integrated Product Development’, PHI
3. John. R. Hartley, Susmu Okamato, ‘Concurrent Engineering, shortening lead times, raising quality & lowering costs’,
4. I. Moustapha, ‘Concurrent Engineering in product Design and Development’, New age International.
5. John Stark, ‘Product Life Cycle Management’, Springer.
6. Michael Grives, ‘Product Lifecycle Management’, Mc Graw Hill.
7. Boothroyd, G., Dewhurst, P., and Knight, W., ‘Product Design for Manufacture and Assembly’, Marcel Dekker Inc.

Course Outcomes: After completion of this course,

CO1: Students will recognize the concept of concurrent engineering and collaborative product development.

CO2: Students will employ the concept of concurrent engineering into product development using Quality Function Deployment (QFD).

CO3: Students will illustrate the product lifecycle management.

CO4: Students will employ quality and into product development.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2								3	3	3	3	2	2

CO2	3	3	3	3		3	2	2		2	3	3	3	3
CO3	2	2	2	3		3	2	3	1	2	2	3	3	3
CO4	3	3	2	3		2		2	2	2	2	3	2	3
AVG	2	2.5	1.75	2.25		2	1	1.75	0.75	2.25	2.5	3	2.75	2.75

PI1606: Micro and Nano Manufacturing

Introduction of micro and nano technology: History and development process of micro cutting, definition and scope of micro cutting, micro and nano-fabrication, concepts of micro and nano-systems and microsystems products.

Micro-cutting mechanics: Material removal at micro-scale: size effect, chip thickness, Microstructure and grain size effects; surface generation and burr formation, tool geometry, tool wear, and tool deflections, tool stiffness and deflections under dynamic loading.

Micro machining, forming, and welding: Diamond micro-machining, micro-turning, micro-milling, micro-drilling, micro-grinding, chemical and electro-chemical micro-machining, electric discharge micro machining, micro and nano structured surface development by nano plastic forming, roller imprinting, micro bending, and micro welding with LASER.

Nano machining and finishing: Focused ion beam machining, Plasma beam machining, Abrasive flow finishing: Magnetic float polishing, Magnetic abrasive finishing, Magneto rheological finishing, Magneto rheological abrasive flow finishing.

Application of micro and nano-system in biomedical: Delivery of diagnostic and therapeutic agents to vascular targets, Real-time biological imaging and detection, Diagnostic and therapeutic applications of metal nano-shells, Future scope of micro and nano system.

Text and References Books:

1. V. K. Jain, ‘Micromanufacturing Processes’, Taylor and Francis, 2012.
2. K. Cheng and D. Hu, ‘Micro-cutting-fundamental and Applications’, John Wiley & Sons, Ltd, 2013.
3. Mark J. Jackson, ‘Micro fabrication & Nano manufacturing’,
4. J. Mc Geough, Marcel Dekker, ‘Micromachining of Engineering Materials’, 2002.
5. N. P. Mahalik, ‘Micromanufacturing & Nanotechnology’, Springer.

Course Outcomes: After completion of this course,

CO1: The course will enable the students to know the basic concepts of and principles of Micro and Nano processes.

CO2: Students will be able to demonstrate the machining mechanics under micro and nano manufacturing processes.

CO3: Students will be able to learn about nano finishing processes and its applications.

CO4: Students will know about advanced application of micro-nano processes in the biomedical.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2

CO1	3	2	1	1	2	1	0	1				3	3	
CO2	3	3	2	3	3	2	0	2				2	3	
CO3	3	1	2	1	3	2	0	2				3	3	
CO4	3	3	3	2	3	3	3	3				3	3	
AVG	3	2.25	2	1.75	2.75	2	0.25	2				2.75	3	

PI1607: Project Management

Introduction: Definition, Classification, Project life cycle phases, stakeholders, project manager;

Defining the Project: Idea generation, Project scope planning and Work Breakdown Structure

Project Scheduling: Development of the network, activity on node and activity on arc network precedence diagrams, Fulkerson's flow algorithm, forward pass and Backward pass computations, slacks and floats, CPM, PERT, Generalized Activity Networks, GERT

Project optimization: Time and Cost trade-off in CPM, crashing procedure, Scheduling when resources are limited. Resource Planning, Resource Allocation, Project Schedule Compression, Algorithms for shortest route problems, minimal spanning tree and maximal flow problems.

Project Performance Measurement: Traditional performance metrics, Earned value analysis

Project Risk Management: Monte Carlo Analysis and Decision Tree Analysis

Project Quality Management: Continuous improvement and Six Sigma tools

Project Closure: Project evaluation, documentation of project.

Project Selection: Project viability, project rating index, NPV and MCDM techniques.

Text and Reference Books:

1. Harold Kerzner, 'Project Management: A Systems Approach to Planning, Scheduling, and Controlling', 9th Edition, John Wiley and Sons, New Jersey, 2003.
2. Prasanna Chandra, 'Projects: Planning, Analysis, Selection, Financing, Implementation, and Review', McGraw Hill Education, Chennai, 2019.
3. J. D. Weist and F. K. Levy, 'A Management Guide to PERT/CPM with GERT/PDM/DCPM and other networks', 2nd edition, Prentice-Hall of India New Delhi, 2003.
4. L. S. Srinath, 'PERT & CPM Principles and Applications', 3rd Edition, East- West Press Pvt. Ltd., New Delhi, 1993.
5. R. G. Ghattas and S. L Mckee, 'Practical Project Management', Pearson Education Asia, 2004.

Course Outcomes: After completion of this course,

CO1: Students will define the project elements to prepare work breakdown structure.

CO2: Students will practice various tools for project scheduling.

CO3: Students will practice resource planning in a project.

CO4: Students will evaluate a project on the basis of various criteria.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	3	2	2		1	2	3	2	1	3
CO2	3	3	3	3	3	1	1		2	2	3	2		3
CO3	3	3	3	3	3	1	2		3	3	3	2	1	3
CO4	3	3	3	3	3	2	3	1	3	3	3	3	1	3
AVG	3	3	3	2.75	3	1.5	2	0.25	2.25	2.5	3	2.25	0.75	3

PI1608: Advanced Materials

Introduction to composite materials: Introduction, classification: polymer matrix composites, metal matrix composites, ceramic matrix composites, carbon–carbon composites, fiber-reinforced composites and nature-made composites, and applications. Reinforcements: Fibers- glass, silica, kevlar, carbon, boron, silicon carbide, and born carbide fibers.

Polymer composites, thermoplastics, thermosetting plastics, manufacturing of PMC, MMC & CCC and their applications. Manufacturing methods: Autoclave, tape production, molding methods, filament winding, hand layup, pultrusion, RTM.

Macro-mechanical analysis of Lamina: Introduction, generalized Hooke's law, reduction of Hooke's law in three dimensions to two dimensions, relationship of compliance and stiffness matrix to engineering elastic constants of an orthotropic lamina, laminate-laminate code.

Functionally graded materials: Types of functionally graded materials-classification- different systems-preparation-properties and applications of functionally graded materials. Shape memory alloys: Introduction-shape memory effect-classification of shape memory alloys-composition-properties and applications of shape memory alloys.

Nano materials: Introduction-properties at nano scales-advantages & disadvantages-applications in comparison with bulk materials (nano structure, wires, tubes, composites). state of art nano advanced- topic delivered by student.

Text and Reference Books:

1. A. K. Bandyopadhyay, 'Nano material', New age Publishers.
2. Robert W.Cahn, 'Material science and Technology: A comprehensive treatment', VCH.
3. Isaac and M Daniel, 'Engineering Mechanics of Composite Materials', Oxford University Press.
4. R. M. Jones, 'Mechanics of Composite Materials', Mc Graw Hill Company, New York, 1975.
5. L. R. Calcote, 'Analysis of Laminated Composite Structures', Van Nostrand Rainfold, NY 1969.
6. B. D. Agarwal and L. J. Broutman, 'Analysis and performance of fibre Composites', Wiley- Inter Science, New York, 1980.
7. Autar K. Kaw, 'Mechanics of Composite Materials', 2nd Edition, CRC Press.

Course Outcomes: After completion of this course,
CO1: To understand the mechanics of different materials.

CO2: To understand the anisotropic material behavior and constituent properties
CO3: Students will be able to Fabrication process of super alloys and different composites.
CO4: Students will be able to design smart and nano materials for engineering applications.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	2	1	0	1				3	2	
CO2	3	3	2	1	2	2	0	2				3	2	
CO3	3	3	3	1	3	1	0	3				3	3	
CO4	3	3	3	1	3	1	2	3				3	3	
AVG	3	3	2.75	1.25	2.25	1.25	0.66	2				3	2.5	

PI1609: Reliability Engineering

Introduction: Concept, certain and impossible events, complementary events, Kolmogorov Axioms, definition of reliability, down time consequences.

Component Reliability Models: Failure data analysis: Failure data, mean failure rate, MTTF, MTBF; Hazard models: Introduction, constant hazard, linearly increasing hazard, The Weibull Model, distributions functions and reliability analysis

System Reliability Models: Conditional probability, multiplication rule, Venn diagram, Bayes' theorem, calculation of system reliability for series, parallel and mixed configuration, logic diagram, k-out-of-m modeling, Markov models- sharing system and Standby system, standby system with switching failure and degrade system.

Reliability Analysis & Allocation: Reliability specification and allocation, failure modes and effects and criticality analysis (FMECA), Element and unit Redundancy, standby redundancy, fault tree analysis, cut sets & tie sets approaches; Reliability estimation- non-parametric, distribution fitting, Replacement policies, monte-carlo simulation, Simple case studies

Maintainability and Availability-Types of Maintenance, Maintainability, Measures of Maintainability, Availability, Life Testing, Quality and Reliability, Introduction to Reliability Availability and Maintainability (RAM), Development of RAM Engineering, Reliability Availability and Maintainability utilization factors, Measurement and Prediction of Human Reliability, Reliability Management

Text and References Books:

1. Charles E Ebling, 'An introduction to reliability and maintainability Engineering', McGraw Hill
2. Patrick D. T. O'Connor, Andre, 'Practical Reliability Engineering', Wiley, 2012.
3. Roy Billinton, Ronald N. Allan, 'Reliability Evaluation of Engineering Systems: Concepts and Techniques', Springer, 1992.
4. K. B. Misra, 'Reliability Analysis and Prediction: A Methodology Oriented Treatment', Elsevier.
5. V. N. A. Naikan, 'Reliability Engineering and Life Testing', PHI, 2008.

6. A. K. Govil, ‘Reliability Engineering’, Tata McGraw Hill New Delhi.
7. E Balagurusamy, ‘Reliability Engineering’, Tata McGraw Hill New Delhi.
8. Brijendra Singh, ‘Quality control and reliability analysis’, Khanna Publication.

Course Outcomes: After completion of this course,
CO1: Students will define reliability, availability, maintainability and associated concepts.
CO2: Students will analyze the component and system reliability of Industrial systems.
CO3: Students will inspect failure modes and estimate reliability
CO4: Students will estimate reliability of humans and industrial systems.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1		2								3		3	1	
CO2	2									3	2	3	3	1
CO3	2			2	2				2	3	2	3	3	3
CO4	2			2	2				2	3	2	3	3	3
AVG	1.5	0.5	0	1	1	0	0	0	1	3	1.5	3	2.5	1.75

PI1610: Work System Design and Ergonomics

Introduction: Productivity, definition and scope of motion and time study, history of motion and time study, work method design, human factor in the application of work study, man – machine interface.

Motion study: concept, operation process chart, flow chart, multiple activity charts, travels chart, flow diagram, and operation analysis. Micro motion study, memo motion study, cycle graphic and chronocyclegraph analysis, fundamental hand motions, therbligs, micro motion study equipment, film analysis- SIMO charts, principles of motion economy.

Time study: concept, uses of time study, time study equipment, making the time study, work sampling, determination of sample size, procedure for selecting random observations, errors in work sampling.

Rating factors and allowances: the concept of qualified worker, the average worker, standard rating and standard performance, definition of rating, systems of rating, rating of efforts, scales of rating, introduction to allowances, classification of allowances, applying the allowances, determining the time standards, predetermining time standards (PTS), standard data, the uses of time standards, MTM-I, MTM-II. Job Evaluation and Merit rating, Wage and Incentive.

Principles of workplace design: physical requirements in the workplace anthropometrics and communication considerations, social requirements of the workplace- personal and territoriality considerations. Workspace design; general principles, deciding position of control with respect to other controls, position of displays with respect to other displays, positioning of displays and controls, control display compatibility.

Ergonomics: Introduction, definition, objectives and scope, man-machine system and its components. Introduction to musculoskeletal system, respiratory and circulatory system, metabolism, measure of physiological functions- workload and energy consumption, Introduction to biomechanics, types of movements of body members, Design of lifting tasks using NIOSH lifting equation, Distal upper extremities risk factors, risk assessment tools; Strain Index, RULA, REBA. Introduction to anthropometry; work table and seat designing. Design of Visual displays and controls. Occupational exposure to; noise, whole body Vibrations, heat stress and dust. Effect of vibration/ noise, temperature, illumination and dust on human health and performance.

Text and Reference Books:

1. R.S. Bridger, ‘Introduction to Ergonomics’, 3rd Edition,
2. Mark S. Sanders, Ernest J. McCormick, ‘Human Factors in Engineering and Design’, McGraw-Hill, Inc.
3. Stephen M. Popkin, Heidi D. Howarth, Donald I. Tepas, ‘Ergonomics of Work Systems’, Wiley.
4. Kim Vincente, ‘The Human Factor’, 1st Edition
5. Marvin E, Mundel & L. David, ‘Motion & Time Study: Improving Productivity’, Pearson.
6. L. P. Singh, ‘Work Study and Ergonomics’, Cambridge University Press.

Course Outcomes: After completion of this course,

CO1: Students will define the concepts of work system design and ergonomics.

CO2: Students will conduct motion study, time study and ration factor evaluation.

CO3: Students will design the workspace based on anthropometric variables.

CO4: Students will study the impact of ergonomic design on the human body.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	3			1	1		2	1	2	1	2	1	2
CO2	1	3	1	3	3	1		3	3	3	3	2	3	3
CO3	3	3	3	3	3	3	2	1	3	3	3	2	3	3
CO4	2	2	1	2	2	3	1		3	3	2	2	3	3
AVG	1.75	2.75	1.25	2	2.25	2	0.75	1.5	2.5	2.75	2.25	2	2.5	2.75

PI1611: Near Net-shape Manufacturing

Introduction to near net-shape operations: Introduction, Classification of near net-shape operations: Sheet metal forming, Massive (bulk)metal forming processes, Injection moulding, Machines for near net-shape operations, Near net-shape operations: past, present and future.

Manufacturing of Near Net Shape Medical Prostheses in Polymeric Sheet by Incremental Sheet Forming: Introduction, Process parameters analysis on polymer basic geometries, Manufacturing of custom cranial implants.

Developments in Friction Stir Processing-A Near Net Shape Forming Technique: Introduction, FSP variables for superplastic forming, Super plastic forming, FSP to modify/repair cast properties.

Evolution in Additive Manufacturing Techniques of Metals as Net-Shaped Products:

Introduction, components and functioning of various additive manufacturing process.

Near Net Shape Manufacturing of Dental Implants Using Additive Processes: Introduction, structure and properties of teeth, requirements and materials for dental implants, additive manufacturing, clinical evaluation of 3D implants.

Laser Additive Manufacturing Processes for Near Net Shape Components: Introduction, Manufacturing of macroscale components, manufacturing of microscale components.

Near Net Shape Manufacturing of Miniature Spur Brass Gears by Abrasive Water Jet Machining: Introduction, Influence of AWJM parameters on surface quality of gears.

Fracture Forming Limits for Near Net Shape Forming of Sheet Metals: Introduction, Failure by facture.

Advances in Near Net Shape Polymer Manufacturing through Microcellular Injection Moulding: Introduction, Theory behind the technology, Enhanced microcellular injection moulding manufacturing techniques.

Text and Reference Books:

1. Kapil Gupta, ‘Near Net Shape Manufacturing Processes’, Springer, 2019.
2. A. Y. C. Nee, S. K. Ong, and Y. G. Wang, ‘Computer applications in Near Net-Shape Operations’, Springer, 1999.

Course Outcomes: After completion of this course,

CO1: To understand the applications of near net shape manufacturing.

CO2: To understand the process chain of near net shape manufacturing processes.

CO3: To analyze the parametric effects in various near net shape manufacturing processes.

CO4: To be able to design for digital manufacturing of different engineering applications.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	3	2	2	1	1	1				3	3	1
CO2	3	3	3	3	3	1	1	1				3	3	1
CO3	3	3	2	3	3	1	1	1				3	2	1
CO4	3	3	2	3	2	1	1	1				2	2	1
AVG	3	2.7	2.5	2.7	2.5	1.0	1.0	1				2.7	2.5	1

Open Elective I (Sixth Semester) **(3-0-0)**

PI1612: Management Information Systems

Basic Concepts of Information System (IS) Role of data and information, Organization structures, Business Process, Systems Approach and introduction to Information Systems.

Types of IS Resources and components of Information System, integration and automation of business functions and developing business models. Role and advantages of Transaction Processing System, Management Information System, Expert Systems and Artificial Intelligence, Executive Support Systems and Strategic Information Systems.

Architecture & Design of IS Architecture, development and maintenance of Information Systems, Centralized and Decentralized Information Systems, Factors of success and failure, value and risk of IS. Decision Making Process Programed and Non-Programed decisions, Decision Support Systems, Models and approaches to DSS.

Introduction to Enterprise Management technologies Business Process Reengineering, Total Quality Management and Enterprise Management System viz. ERP, SCM, CRM and Ecommerce.

Introduction to SAD System Analysis and Design. Models and Approaches of Systems Development.

Text and Reference Books:

1. O. Z. Effy, 'Management Information Systems', Thomson Leaning and Vikas Publications.
2. James A. O'Brein, 'Management Information Systems', Tata McGraw-Hill.
3. W.S Jawadekar, 'Management Information System', Tata Mc Graw Hill Publication.
4. David Kroenke, 'Management Information System', Tata Mc Graw Hill Publication.
5. D.P. Goyal, 'MIS: Management Perspective', Macmillan Business Books.
6. Raj K. Wadwha, Jimmy Dawar, P. Bhaskara Rao, 'MIS and Corporate Communications', Kanishka Publishers.
7. Kenneth C. Landon, Jane P. Landon, 'MIS: Managing the digital firm', Pearson Education.

Course Outcomes: After completion of this course,

CO1: Students will recognize the concepts and role of management information system.

CO2: Students will classify the information system resources and components.

CO3: Students will define the information system architecture.

CO4: Students will demonstrate modern technologies in MIS.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1						3	2	2		3	3	3	1	1
CO2						3	2	2		3	2	3	2	3
CO3						3	1	2		3	2	3	3	3
CO4						3	3	2		3		2	1	2
AVG						3	2	2		3	1.75	2.75	1.75	2.25

PI1613: Total Quality Management

Introduction: Definition of quality, Dimensions of quality, Evolution of quality, Quality control (QC), Quality assurance (QA), Quality planning (QP), Fundamentals of total quality management (TQM), Some important philosophies and their impact on quality (Deming, Juran, Crossby), Identification and measurement of quality costs, Issues related to products, processes, organization, leadership and commitment for total quality achievement.

TQM Principles and Strategies: Customer satisfaction, Employee involvement, Continuous process improvement, Supplier partnership, Performance measures.

TQM Tools: Business process benchmarking (BPB), Quality function deployment (QFD), Failure mode and effect analysis (FMEA), Six sigma - Basic Concept, Methodology, Process Improvement Model (DMAIC) Steps (Objectives, Tools and Techniques used), Six sigma organization and its Implementation requirements, Introduction to lean manufacturing, JIT, Kaizen, Total productive maintenance

Statistical Process Control Techniques: Definition of statistical quality control (SQC), Benefits and limitations of SQC, Introduction to statistics, Seven quality control tools (old and new), Control charts.

Quality Systems: Need for ISO 9000 and Other Quality Systems, ISO 9000:2000 Quality System – Elements, Implementation of Quality System, Documentation, Quality Auditing, TS 16949, ISO 14000 – Concept, Requirements and Benefits.

Text and Reference Books:

1. Charantimath, Poornima M., ‘Total Quality Management’, Pearson.
2. Ramasamy S, ‘Total Quality Management’, McGraw-Hill.
3. Mitra A, ‘Fundamental of Quality Control and Improvement’, Wiley.

Course Outcomes: After completion of this course,
CO1: The students will identify the dimensions of product or service quality and classify the costs associated with quality.
CO2: The students will apply TQM tools for identifying and solving quality related problems.
CO3: The students will apply the statistical concepts for quality control of products and services.
CO4: The students will recognize different quality systems.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1			3	2	1	2	3	3	3	1	1
CO2	3	3	3	3	3	3	2	1	3	3	2	3	2	3
CO3	3	3	3	3	3	3	1	1	3	3	2	3	3	3
CO4	3	3	2	1	1	3	3	2	2	3		2	1	2
AVG	3	3	2.25	1.75	1.75	3	2	1.25	2.5	3	1.75	2.75	1.75	2.25

PI1614: Business Statistics

Frequency Distribution and Graphs: Frequency, Stem and Leaf Display, Frequency Distributions, Data Grouping: Discrete and Continuous, Introduction to Graphs, Graph for Qualitative variables, Graph for Quantitative variables, Various types of graphs and diagrams: pictographs, bar diagram, scatter diagram, histogram, pie chart, frequency curve and frequency polygon.

Measures of Central Tendency and Dispersion: Mean, Median and Mode, Weighted Average, Geometric Mean, Harmonic Mean, Relative merits of Mean, Median and Mode in a distribution, Mean of two or more means, Measures of Dispersion, Range, Co-efficient of

Range, Quartiles, Inter-Quartile Range and Quartile Deviation, Coefficient of Quartile Deviation, Mean Deviation, Coefficient of Mean Deviation, Standard Deviation, Coefficient of Variation, Skewness and Kurtosis; Measures of Skewness: Absolute and Relative; Co-efficient of Skewness: Karl Pearson's, Bowley's and Kelly's; Moments and Moments based measures of Skewness ($\beta 1$)and Kurtosis ($\beta 2$).

Theoretical Probability Distributions: Introduction - Sample space and Events, Simple and Compound Events, Probability and Probability distributions: Normal distribution, Binomial and Poisson Distribution.

Sampling and Sampling Distributions: Introduction , Population and Sample - Universe or Population - Types of Population – Sample , Advantages of Sampling, Sampling Theory - Law of Statistical Regularity - Principle of Inertia of Large Numbers - Principle of Persistence of Small Numbers - Principle of Validity - Principle of Optimization , Terms Used in Sampling Theory , Errors in Statistics , Measures of Statistical Errors , Types of Sampling - Probability Sampling - Non-Probability Sampling, Case let on Types of Sampling, Determination of Sample Size, Central Limit Theorem.

Estimation: Introduction, Reasons for Making Estimates , Making Statistical Inference, Types of Estimates - Point estimate - Interval estimate , Criteria of a Good Estimator – Unbiasedness – Efficiency – Consistency – Sufficiency, Point Estimates, Interval Estimates, Case study on calculating estimates - Making the interval estimate, Interval Estimates and Confidence Intervals - Interval estimates of the mean of large samples - Interval estimates of the proportion of large samples - Interval estimates using the Student's 't' distribution , Determining the Sample Size in Estimation

Text and Reference Books:

1. Roger E. Kirk,' Statistics: An Introduction', 5th Edition, Thomson-Wadsworth Publication.
2. M. L. Berenson and D. M. Levine D.M., 'Basic Business Statistics', Prentice-Hall, Englewood Cliffs, New Jersey, 1996.
3. Bhattacharyya, G. K., and R. A. Johnson, 'Statistical Concepts and Methods', John Wiley and Sons, New York,1997.
4. Charles Wheelan, 'Naked Statistics: Stripping the Dread From the Data', 1st Edition, W. W. Norton & Company, 2013.
5. Mc Clave, Benson and Sincich, 'Statistics for Business and Economics', 11th Edition, Prentice Hall Publication.
6. Goon, A.M., Gupta, M.K. and Dasgupta, B., 'An Outline of Statistical Theory', Vol. I, 4th Edition. World Press, Kolkata, 2003.
7. Rohatgi V. K. and Saleh, A.K. Md. E., 'An Introduction to Probability and Statistics', 2nd Edition, John Wiley and Sons, 2009.
8. Richars I. Levin, David S. Rubin, Sanjay Rastogi, Masood Husain Siddiqui, 'Statistics for Management', 7th Edition, Pearson Education.
9. S. P. Gupta, 'Statistical Methods', S. Chand Publication.

Course Outcomes: After completion of this course,

CO1: The students will illustrate appropriate graphical and numerical descriptive statistics for different types of data.

CO2: The students will apply probability rules and concepts relating to discrete and continuous random variables to answer questions within a business context.

CO3: The students will demonstrate knowledge of the importance of the Central Limit Theorem (CLT) and its applications.

CO4: The students will apply simple/multiple regression models to analyse the underlying relationships between the variables through hypothesis testing to aid decision making in a business context.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	3	2				1	2	1	1	1	2
CO2	3	3	3	3	3				2	2	3	3	1	3
CO3	3	3	3	3	2				1	2	1	1	1	2
CO4	3	3	3	3	2	1			1	2	3	3	1	3
AVG	3	3	3	3	2.25	0.25			1.25	2	2	2	1	2.5

PI1615: Additive Manufacturing

Introduction to Additive Manufacturing: Introduction to AM, AM evolution, Distinction between AM & CNC machining, Steps in AM, Classification of AM processes, Advantages of AM and Types of materials for AM. Part Orientation and Support Structure Generation, Model Slicing and Contour Data Organization, Direct and Adaptive Slicing, Hatching Strategies and Tool Path Generation.

Vat Photopolymerization AM Processes: Stereolithography (SL), Materials, Process Modeling, SL resin curing process, SL scan patterns, Micro-stereolithography, Mask Projection Processes, Two-Photon vat photopolymerization, Process Benefits and Drawbacks, Applications of Vat Photopolymerization, Material Jetting and Binder Jetting AM Processes.

Extrusion-Based AM Processes: Fused Deposition Modelling (FDM), Principles, Materials, Process Modelling, Plotting and path control, Bio-Extrusion, Contour Crafting, Process Benefits and Drawbacks, Applications of Extrusion-Based Processes.

Sheet Lamination AM Processes: Bonding Mechanisms, Materials, Laminated Object Manufacturing (LOM), Ultrasonic Consolidation (UC), Gluing, Thermal bonding, LOM and UC applications.

Powder Bed Fusion AM Processes: Selective laser Sintering (SLS), Materials, Powder fusion mechanism and powder handling, Process Modelling, SLS Metal and ceramic part creation, Electron Beam melting (EBM), Process Benefits and Drawbacks, Applications of Powder Bed Fusion Processes.

Directed Energy Deposition AM Processes: Process Description, Material Delivery, Laser Engineered Net Shaping (LENS), Direct Metal Deposition (DMD), Electron Beam Based Metal Deposition, Processing-structure-properties, relationships, Benefits and drawbacks, Applications of Directed Energy Deposition Processes.

Text and Reference Books:

1. Ian Gibson, David W Rosen, Brent Stucker, 'Additive Manufacturing Technologies: 3D Printing, Rapid Prototyping, and Direct Digital Manufacturing', 2nd Edition, Springer, 2015.
2. Martin Leary, 'Design for Additive Manufacturing', 1st Edition, Elsevier, 2019.
3. Chee Kai Chua and Kah Fai, '3D Printing and Additive Manufacturing: Principles and Applications', 5th Edition, World Scientific, 2017.
4. Milewski, John O., 'Additive Manufacturing of Metals', 1st Edition, Springer International Publishing, 2017.
5. Chua Chee Kai, Leong Kah Fai, 'Rapid Prototyping: Principles & Applications', World Scientific, 2003.
6. Ali K. Kamrani, Emand A. Nasr, 'Rapid Prototyping: Theory & Practice', Springer, 2006.
7. D.T. Pham, S.S. Dimov, 'Rapid Manufacturing: The Technologies and Applications of Rapid Prototyping and Rapid Tooling', Springer, 2001.

Course Outcomes: After completion of this course,
CO1: To understand the need of additive manufacturing and classification of additive manufacturing processes
CO2: To understand the process chain of additive manufacturing processes
CO3: To analyze the parametric effects in various additive manufacturing processes
CO4: To be able to design for additive manufacturing of different engineering applications

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	2	1	1	1	1			3	3	1
CO2	3	3	3	3	3	1	1	1	2			3	3	1
CO3	3	3	3	3	3	1	1	1	1			3	2	1
CO4	3	3	2	3	2	1	1	1	1			3	3	1
AVG	3	3	2.7	2.7	2.5	1.0	1.0	1	1.25			3	2.7	1

IV Year: I-Semester [7th semester]

PI1701: Automation and Robotics **(3-0-0)**

Introduction to Industrial Automation and Control, Architecture of Industrial Automation Systems, Introduction to sensors and measurement systems, Temperature measurement, Pressure and Force measurements, Displacement and speed measurement, Flow measurement techniques, Measurement of level, humidity, pH etc, Signal Conditioning and Processing, Estimation of errors and Calibration.

Control of Machine tools: Introduction to CNC Machines, Analysis of a control loop, Introduction to Actuators: Flow Control Valves, Hydraulic Actuator Systems: Pumps and Motors, Proportional and Servo Valves, Pneumatic Control Systems: System Components, Pneumatic Control Systems: Controllers and Integrated Control Systems.

Robotic Systems: Fundamentals of robotics and its technology, robot classification based on geometry, devices, control and path movement, robot motion analysis, robot selection and its application, economic justification of robots.

Concept of Robotic/Machine vision, Teach pendant, Industrial Applications of Robots for material transfer, machine loading / unloading, welding, assembly and spray painting operations.

Anatomy and structural design of robot, Manipulation and geometry, Robot control systems and components, Interlocks and sequence control, Robot and effectors, Robot drive system, Robot programming.

Numerical Control: Conventional numerical control (NC) – basic components of an NC system, applications of NC, economics of NC, and problems with conventional NC. Computer Numerical Control (CNC), Direct Numerical Control (DNC), and combined CNC/DNC systems. NC programming.

Text and Reference Books:

1. M. P. Groover, M. Weiss, R. N. Nagel, N. G. Odrey, 'Industrial Robotics-Technology, Programming and applications', McGraw hill, New Delhi.
2. M. P. Groover, 'Automation, Production systems and computer, Integrated manufacturing', Prentise hall of India Pvt. L, New Delhi.

Course Outcomes: After completion of this course,

CO1: To understand the different types and labels of automation in industrial manufacturing processes

CO2: To understand the different types of control systems used in manufacturing and industries

CO3: To analyze the robotics systems and types of robots

CO4: To be able to design manufacturing and assembly for products based on automation through CNC and robots

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	3	2	1	1					3	3	1
CO2	3	3	3	3	3	1	1					3	3	1
CO3	3	2	3	3	3	1	1					3	2	1
CO4	3	3	2	3	3	1	1					3	3	1
AVG	3	2.7	2.7	3	2.7	1.0	1.0					3	2.7	1

PI1702: Introduction to Machine Learning (3-0-0)

Probability Theory, Linear Algebra, Convex Optimization, Introduction: Statistical Decision Theory - Regression, Classification, Bias Variance, Linear Regression, Multivariate Regression, Subset Selection, Shrinkage Methods, Principal Component Regression, Partial Least squares Linear Classification, Logistic Regression, Linear Discriminant Analysis

Perceptron, Support Vector Machines, Neural Networks - Introduction, Early Models, Perceptron Learning, Backpropagation, Initialization, Training & Validation, Parameter Estimation - MLE, MAP, Bayesian Estimation

Decision Trees, Regression Trees, Stopping Criterion & Pruning loss functions, Categorical Attributes, Multiway Splits, Missing Values, Decision Trees - Instability Evaluation Measures, Bootstrapping & Cross Validation, Class Evaluation Measures, ROC curve, MDL, Ensemble Methods - Bagging, Committee Machines and Stacking, Boosting

Gradient Boosting, Random Forests, Multi-class Classification, Naive Bayes, Bayesian Networks, Undirected Graphical Models, HMM, Variable Elimination, Belief Propagation, Partitional Clustering, Hierarchical Clustering, Birch Algorithm, CURE Algorithm, Density-based Clustering Gaussian Mixture Models, Expectation Maximization

Text and References Books:

1. Nils J.Nilsson, ‘Introduction to Machine learning’,
2. Judith Hurwitz and Daniel Kirsch, ‘Machine learning for dummies’, IBM Limited.
3. Andreas, C. Muller & Sarah Guido, O'Reilly, ‘Introduction to Machine Learning with Python A guide for data scientists’,
4. Trevor Hastie, Robert Tibshirani, Jerome H. Friedman, ‘The Elements of Statistical Learning’,

Course Outcomes: After completion of this course,

CO1: Students will develop an appreciation for what is involved in Learning models from data.

CO2: Students will apply a wide variety of learning algorithms

CO3: Students will evaluate models generated from data.

CO4: Students will apply the algorithms to a real problem, optimize the models learned and report on the expected accuracy that can be achieved by applying the models

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	3	2	1			1		3	3		3
CO2	3	3	3	3	3	1			1		3	3		3
CO3	3	2	3	3	3	1			1		3	3		2
CO4	3	3	2	3	3	1			1		3	3		3
AVG	3	2.7	2.7	3	2.7	1.0			1		3	3		2.7

Professional Elective III & IV (Seventh Semester) (3-0-0)

PI1703: Characterization of Engineering Materials

Fundamentals of optics, Construction of optical microscope and its working principle, Image formation in optical microscope, Sample preparation for optical microscopy

Introduction to scanning electron microscopy (SEM), Construction and working principle of

SEM, Image formation in SEM, Sample preparation for SEM, Applications of SEM, Elemental analysis using energy dispersive spectroscopy (EDS)

Fundamentals of X-ray diffraction, Bragg's law, Crystallite size, Peak shift, Peak broadening due to strain and crystallite size

Introduction to transmission electron microscopy (TEM), Construction and working principle of TEM, Image formation in TEM, Sample preparation for TEM, Applications of TEM

Assessment of hardness through Brinell hardness test, Rockwell hardness test, Vickers's microhardness test, Nanoindentation test, Assessment of Young's Modulus through nanoindentation test, Assessment of scratch resistance through macroscopic and nano-scratch tests.

Tribological characterization through pin-on-disc test

Electrochemical characterization through potentiodynamic-polarization test (PDP)

Text and Reference Books:

1. Douglas B. Murphy, 'Fundamentals of light microscopy and electronic imaging', Wiley-Liss, Inc. USA.
2. C. Richard Brundle, Charles A. Evans, Jr., Shaun Wilson, 'Encyclopedia of Materials Characterization, Surfaces, Interfaces, Thin Films', Butterworth-Heinemann, Boston, US.
3. B. D. Cullity and S. R. Stock, 'Elements of X-ray diffraction', Prentice Hall, Inc., USA.
4. D.B. Williams and C. Barry Carter, 'Transmission electron microscopy', 4 Vol., Springer, USA.
5. 'ASM Handbook: Materials Characterization', ASM International, 2008.
6. Prof. S. Sankaran, 'Materials Characterization', IIT Madras (NPTEL Course)
7. Prof. Shibayan Roy, 'Techniques of Material Characterization', IIT Kharagpur, (NPTEL Course)
8. Prof. A. K. Chattopadhyay, 'Technology of Surface Coating', IIT Kharagpur (NPTEL Course)
9. Prof. Tamal Banerjee, 'Physical and Electrochemical Characterizations in Chemical Engineering', IIT Guwahati (NPTEL Course)

Course Outcomes: After completion of this course,
CO1: Students will be able to learn different microscopic techniques
CO2: Students will be able to learn and apply concepts of mechanical characterization
CO3: Students will be able to assess the tribological properties of materials
CO4: Students will be able to evaluate the electrochemical properties of materials

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	0	3		0	3				3	3	
CO2	3	2	1	0	3		0	2				3	3	
CO3	3	3	2	1	2		2	2				3	3	
CO4	3	3	2	1	2		2	2				3	3	

AVG	3	2.5	1.5	0.66	2.5		1	2.25				3	3	

PI 1704: Design for Manufacturing and Assembly

Introduction to Design for Manufacturing and Assembly (DFMA): Engineering design process and its structure, steps in design process, traditional design methods, Introduction to DFMA: History of DFMA, Steps for applying DFMA during product design, Advantages of applying DFMA during product design, Reasons for not implementing DFMA.

Design for Manual Assembly: Introduction, General Design Guidelines for Manual Assembly, Design Guidelines for Part Handling, Design Guidelines for Insertion and Fastening, Development of a Systematic Design for Assembly (DFA) Analysis Method, DFA Index, Classification System for Manual Handling, Classification System for Manual Insertion and Fastening, Effect of Part Symmetry on Handling Time, Effect of Part Thickness and Size on Handling Time, Effect of Weight on Handling Time, Parts Requiring Two Hands for Manipulation, Threaded Fasteners, Effects of Holding Down, Problems with Manual Assembly Time Standards, Application of the DFA Method.

Design for die casting: Die casting materials and machines, Trimming dies, optimum number of cavities, determination of machine size, die casting cycle time, cost estimation, design principles.

Design for manufacturing of sheet metal parts: Manufacturing cost estimation using individual dies like, shearing die, piercing die, bending die and deep drawing dies, design principles.

Design for Powder metallurgical parts: Introduction to powder metallurgy, powder metallurgy tooling, selection of appropriate press, profile complexity of powder metal parts, calculation of manufacturing costs.

Text and Reference Books:

1. Geoffrey Boothroyd, Peter Dewhurst and Winston Knight, ‘Product Design for Manufacture and Assembly’, 2nd Edition, CRC press, Taylor & Francis, Florida, USA.
2. O. Molloy, S. Tilley and E. A. Warman, ‘Design for Manufacturing and assembly’, 1st Edition, Chapman &Hall, London, UK.
3. A. K. Chitale and R. C. Gupta, ‘Product design and Manufacturing’, Prentice Hall of India, New Delhi.
4. Geoffrey Boothroyd, ‘Assembly Automation and Product Design’, 2nd Edition, CRC press, Taylor & Francis, Florida, USA.

Course Outcomes: After completion of this course,

CO1: Students will recognize the role of DFMA in product design and development.

CO2: Students will propose the product design with reduced assembly time in manual assembly method.

CO3: Students will analyse the product design to reduce the manufacturing hours.

CO4: Students will propose the product design with reduced assembly time in automatic and robotic assembly methods.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	3		2		1	1			3	3	2
CO2	3	3	3	3		2		1	1			3	3	2
CO3	3	3	3	2		2		1	1			3	3	2
CO4	3	3	3	3		2		1	1			3	3	2
AVG	3	3	3	2.75		2		1	1			3	3	2

PI1705: Value Engineering

Introduction: Value engineering concepts, advantages, applications, problem recognition, and role in productivity, criteria for comparison, element of choice. Organization: Level of value engineering in the organization, size and skill of Value Engineering staff, Value engineering team, VE activity, unique and quantitative evaluation of ideas.

Value Engineering Job Plan: Introduction, orientation, information phase, speculation phase, analysis phase. Selection and Evaluation of value engineering Projects, Project selection, methods selection, value standards, application of value engineering methodology. Analysis Function: Anatomy of the function, use esteem and exchange values, basic vs. secondary vs. unnecessary functions. Approach of function, Evaluation of function, determining function, classifying function, evaluation of costs, evaluation of worth, determining worth, evaluation of value.

Value Engineering Techniques: Selecting products and operation for value engineering action, value engineering Programs, determining and evaluating function(s) assigning rupee equivalents, developing alternate means to required functions, decision making for optimum alternative, use of decision matrix, queuing theory and Monte Carlo method make or buy, measuring profits, reporting results, Follow up, Use of advanced technique like Function Analysis System.

Value Engineering in Product Design: Fast diagramming: cost models, life cycle costs, Case Studies and roadblocks, Value engineering operation in maintenance and repair activities, value engineering in non-hardware projects

Text and Reference Books:

1. Anil Kumar Mukhopadhyaya, ‘Value Engineering: Concepts Techniques and applications’, SAGE Publications.
2. S. S. Iyer, ‘Value Engineering a How-to Manual’, New Age International Publishers, New Delhi
3. O.P. Khan, ‘Industrial Engineering & Management’, Dhanpat Rai & Sons.

Course Outcomes: After completion of this course,
CO1: The students will relate value engineering to costs, process, machinery and price and its application to decision making.
CO2: The students will use value engineering as an economic analysis tool.
CO3: The students will appraise the value engineering operation in maintenance and repair activities.

CO4: The students will create the value engineering team and discuss the value engineering case studies

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	3				3		2		2		3	2	1
CO2	2	3	1	2	3	2	2	2	2	2	2	3	2	1
CO3	3	3	2	3	2		2	1	2	1	3	2	2	3
CO4	3	3	3	2		1			3	3	3	2	2	3
AVG	2.5	3	1.5	1.75	1.25	1.5	1	1.25	1.75	2	2	2.5	2	2

PI 1706: Modeling and Simulation

Introduction: Concept of system and environment, components of a system, discrete and continuous systems, linear and nonlinear systems, stochastic activities, areas of applications of simulation

Physical modeling: Models of a system, types of models, stochastic and dynamic models, principles used in modeling, guidelines for determining the level of model details, techniques for increasing model validity and credibility

Probability concepts: Random variables and their properties, discrete and continuous probability functions, Poisson processes, empirical distributions, random numbers, random numbers and random variate generation.

System simulation: Techniques of simulation, Monte Carlo method, experimental nature of simulation, numerical computation techniques, output data analysis for single system, comparing alternative system configurations, statistical procedure for comparing real world observations with simulation data.

Simulation of queuing systems: Introduction, components of a waiting line system, stationary and time dependent queues, costs of customer waiting time and idle capacity, time flow mechanism, long run measure of performance of queuing systems, study state behavior, of finite and infinite population models.

Design of simulation experiments: Verification and validation of simulation, validation of models, design of simulation experiment, length of simulation run, replication of runs, batch means, elimination of initial bias, statistical independence of observations, variance reduction techniques,

Case studies: Simulation of various inventory systems, simulation of PERT, simulation of manufacturing and material handling systems with softwares.

Text and Reference Books:

1. J. banks, John S Carson, ‘Discrete Event System Simulation’, Prentice Hall Inc.
2. Averill M Law, and W. D. Kelton, ‘Simulation Modeling and Analysis’, McGraw Hill Inc.
3. J. Schwarzenbach, and K. F. Gill, ‘System Modeling and Control’, Edward Arnold.
4. Allan Carrie, ‘Simulation of Manufacturing Systems’, John Wiley and Sons.
5. Viswanadhan, and Narahari, ‘Performance modeling of Automated Manufacturing

Systems’, Prentice Hall Inc.,

Course Outcomes: After completion of this course,
CO1: The students will develop manufacturing models of discrete event systems.
CO2: The students will generate and analyze uncertainty using random numbers and random variates.
CO3: The students will simulate systems to compare alternatives and find optimal solutions to industrial problems.
CO4: The students will design simulation experiments and, verify and validate models using input output analysis.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	3	2	3	1				2	2	3	2	2	3
CO2	1	2	2	3	3				2	2	3	2	2	3
CO3	3	3	3	3	3				3	2	3	2	2	3
CO4	3	3	3	3	3				3	2	3	2	2	3
AVG	2.25	2.75	2.5	3	2.5				2.5	2	3	2	2	3

PI1707: Finite Element Method

Introduction to Finite Element methods – General descriptions, brief history of development, advantages, disadvantages, basic steps of FEM, limitations, error and accuracy in FEM.

Introduction, Functional, Principle of Virtual Work, Principle of Minimum Potential Energy, Ritz method, Galerkin method, Least squares, collocation and subdomain methods, Examples for Ritz and Galerkin methods, Ritz FEM formulation, Galerkin FEM formulation

Introduction, Finite element formulation for 1-D problems, Coordinate and shape functions, Elemental matrices, assembly, elimination method, penalty method, multipoint constraints. Implementation of FEM for plane and 3-D Trusses examples.

Introduction, Finite element formulation for 2-D problems, constant strain triangle, various elements, ISO parametric, sub parametric and super parametric elements, interpolation functions, numerical integration, two dimensional integrals, plane stress, plane strain, axisymmetric, plate bending problems.

Heat conduction problem: 2-D heat conduction analysis, formulation of functional, element matrices and case studies. Fluid flow problem: Stream function formulation, velocity potential formulation and torsional analysis of a prismatic bar.

Finite element formulation for 3-D problems, mesh, preparation, hexahedral elements and higher order element, Problem modeling, Applications of finite element modeling of manufacturing processes in casting, forming, welding and machining, etc.

Text and Reference Books:

1. P. Seshu, ‘Textbook of Finite Element Analysis’, PHI, 2009.
2. U. S. Dixit, ‘Finite Element Methods for Engineers’, Cengage Learning, New Delhi, 2009.

3. J. N. Reddy, 'Finite Element Method in Engineering', Tata McGraw Hill, 2007.
4. S. S. Rao, 'The finite element method in engineering', Butterworth-Heinemann, 2017.
5. Cook, D. Robert, 'Concepts and applications of finite element analysis', John Wiley & Sons, 2007.
6. K. J. Bathe, 'Finite element procedures', Prentice Hall, New Jersey, 2006.

Course Outcomes: After completion of this course,

CO1: Understand the concept and elementary procedure of FEM

CO2: Apply FEM to one dimensional structure like trusses, beams and frames.

CO3: Analyze plain stress, plain strain, and axi-symmetric and plate bending problems.

CO4: Formulate the FE equations for heat transfer and fluid flow problem.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	3	3	3							3	3	1
CO2	3	3	2	3	3							3	3	1
CO3	3	3	3	2	3							3	2	1
CO4	3	3	3	2	3							2	2	1
AVG	3	2.7	2.7	2.5	3							2.7	2.5	1

PI1708: Lean Manufacturing

Introduction Lean Manufacturing: Seven forms of waste and their description, Historical evolution of lean manufacturing, Global competition, Customer requirements, Requirements of other stake holders, Meaning of Lean Manufacturing System (LMS), Meaning of Value and waste, Need for LMS, Symptoms of underperforming organizations, Meeting the customer requirement, Elements of LMS. Traditional Vs lean manufacturing

Primary Tools of Lean manufacturing: 5-S, Workplace organization, Total Productive Maintenance, Process mapping/ Value stream mapping, Work cell.

Secondary Tools of Lean manufacturing: Objective and benefits of Secondary lean tool, Cause and Effect diagram, Pareto chart, Spider chart, Poka yoke, Kanban, Automation, Single minute exchange of die (SMED), Design for manufacturing and assembly, value stream mapping, Just in time (JIT), Visual workplace, TQM Tools And Techniques, Lean with ISO 9001:2000.

Total Productive Maintenance: Objectives and functions, Tero technology, Reliability Centered Maintenance, maintainability prediction, availability and system effectiveness, maintenance costs, maintenance organization. Minimal repair, maintenance types, Predetermined Maintenance, Preventive Maintenance, Corrective Maintenance, Condition-based Maintenance, Predictive Maintenance and Reactive Maintenance, Primary and secondary tool for TPM, Case studies related to TPM.

Text and Reference Books:

1. Yasuhiro Monden, 'Toyota Production System: An Integrated Approach to Just-In-Time', 4th Edition, CRC Press, 2011.

2. A. Mitra, 'Fundamentals of Quality Control and Improvement', 2nd Edition, PHI, 1998.
3. J. Evans and W. Linsay, 'The Management and Control of Quality', 6th Edition, Thomson, 2005.
4. D. H. Besterfield, 'Total Quality Management', 3rd Edition, Pearson Education, 2008.
5. D. C. Montgomery, 'Design and Analysis of Experiments', John Wiley & Son
6. N. Goplakrishnan, 'Simplifed Lean Manufacture', PHI, 2010.
7. Pascal Dennis, 'Lean Production Simplified', Productivity Press, 2007.

Course Outcomes: After completion of this course,
CO1: Students will understand the need for lean manufacturing systems.
CO2: Students will explain the approaches to, concepts, and theories of lean manufacturing.
CO3: Students will apply lean tools and techniques in the efforts to introduce lean practices within the organization to effect continuous improvement.
CO4: Students will determine the maintenance schedules in plants.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	2	1	2		2	3	2	1	2	2	3	1	1
CO2	3	3	3	3	3	3	3		3	3	3	3	2	3
CO3	3	3	3	3	3	3	3	2	3	3	2	3	3	3
CO4	3	3	2	1	2	1	2		2	3	2	2	1	2
AVG	2.5	2.75	2.25	2.25	2.25	2.25	2.75	1	2.25	2.75	2.25	2.75	1.75	2.25

PI1709: Safety in Engineering Industry

Safety in metal working machinery and wood working machines: General safety rules, principles, maintenance, Inspections of turning machines, boring machines, milling machine, planning machine and grinding machines, CNC machines, safety principles, electrical guards, work area, material handling, inspection, standards and codes- saws, types, hazards.

Principles of machine guarding: Guarding during maintenance, Zero Mechanical State (ZMS), Definition, Policy for ZMS – guarding of hazards - point of operation protective devices, machine guarding, types, fixed guard, interlock guard, automatic guard, trip guard, electron eye, positional control guard, fixed guard fencing- guard construction- guard opening.

Safety in welding and gas cutting: Gas welding and oxygen cutting, resistances welding, arc welding and cutting, common hazards, personal protective equipment, training, safety precautions in brazing, soldering and metalizing – explosive welding, selection, care and maintenance of the associated equipment and instruments – safety in generation, distribution and handling of industrial gases-colour coding – flashback arrestor – leak detection-pipe line safety-storage and handling of gas cylinders.

Safety in cold forming and hot working of metals: Cold working, power presses, point of operation safe guarding, auxiliary mechanisms, feeding and cutting mechanism, hand or foot-operated presses, power press electric controls, power press set up and die removal, inspection and maintenance-metal sheers-press brakes. Hot working safety in forging, hot rolling mill operation, safe guards in hot rolling mills – hot bending of pipes, hazards and control measures.

Safety in gas furnace operation, cupola, crucibles, ovens, foundry health hazards, work environment, material handling in foundries, foundry production cleaning and finishing foundry processes

Safety in finishing, inspection and testing: Heat treatment operations, electro plating, paint shops, sand and shot blasting, safety in inspection and testing, dynamic balancing, hydro testing, valves, boiler drums and headers, pressure vessels, air leak test, steam testing, safety in radiography, personal monitoring devices, radiation hazards, engineering and administrative controls, Indian Boilers Regulation. Health and welfare measures in engineering industry-pollution control in engineering industry industrial waste disposal.

Text and Reference Books:

1. 'Accident Prevention Manual for Industrial Operation', 19th Edition, Chicago, III.: National Safety Council, 1982.
2. 'Occupational safety Manual', BHEL, Trichy, 1988.
3. John V. Grimaldi and Rollin H. Simonds, 'Safety Management', All India Travelers Book seller, New Delhi, 1989.
4. N.V. Krishnan, 'Safety in Industry', Jaico Publisher House, 1996.
5. 'Indian Boiler acts and Regulations', Government of India.
6. 'Safety in the use of wood working machines', HMSO, UK, 1992.
7. 'Health and Safety in welding and Allied processes', welding Institute, UK, High Tech. Publishing Ltd., London, 1989.

Course Outcomes: After completion of this course,

CO1: Students will recognize the rules and principles of safety for various machines.

CO2: Students will demonstrate the machine guarding during maintenance.

CO3: Students will practice safety in welding, gas cutting, forming and melting of metals.

CO4: Students will practice safety in finishing and treatment processes.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2				3	1		3			3	3	1
CO2	2	2				3	2		3			3	3	1
CO3	2	2				3	3		3			3	2	1
CO4	2	2				3	3		3			2	2	1
AVG	2	2				3	2.25		3			2.7	2.5	1

PI1710: Integrated Computational Materials Engineering

Principles of Materials Engineering, Introduction to Integrated Computational Materials Engineering (ICME); Overview of ICME and history. Multiscale Aspects of Materials, Introduction to ICME Techniques and Tools

Thermodynamics of Materials Engineering, Computer Simulations at Different Time Scales, Finite Element Modeling and Computational Solid Mechanics, Computational Thermodynamics and Kinetics of Materials.

Electronic Structure Theory and Methods, Introduction of First-Principles Methods, Electronic Structure Calculations - their applications in materials design

Atomistic Modeling and Simulations of Materials – fundamentals and applications of kinetic Monte Carlo, molecular statics, and molecular dynamics simulations.

Mesoscale Modeling of Process-Structure Relations in Materials - Phase-field modeling, Cellular Automata, Computational Micromechanics - Discrete Dislocation Dynamics and Crystal Plasticity, Lectures.

Materials Process Modeling; Applications of Computational Fluid Dynamics – Modeling of Casting, welding and additive processes, Examples related to microstructure evolution and modeling.

Optimization and Machine Learning in Materials Science – data-driven modeling of process-structure-property-performance relations, Concurrent and Parallel Programming, Information and Tools Integration for ICME.

Text and Reference Books:

1. Mark F. Horstemeyer, 'Integrated Computational Materials Engineering for Metals: Concepts and Case Studies', John Wiley & Sons, Inc, 2018.
2. Somnath Ghosh, Christopher Woodward, Craig Przybyla, 'Integrated Computational Materials Engineering: Advancing Computational and Experimental Methods', Springer, 2020.

Course Outcomes: After completion of this course,

CO1: acquire knowledge of structure-property relations of different engineering materials and processing, and multi-scale modeling techniques.

CO2: To be able to understand and use different mathematical tools needed for quantitative characterization of microstructure and calculation of effective properties.

CO3: Have the knowledge of the various tools and techniques needed to characterize and design heterogeneous materials, and able to solve practical problems

CO4: Apply the knowledge of ICME tools for integrated product process design for different engineering materials.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	3	3	1	1					3	3	2
CO2	3	3	2	3	3	1	1					2	2	2
CO3	3	3	3	3	3	1	1					3	3	2
CO4	2	2				3	3		3			2	2	1
AVG	3	3	2.7	3	3	1	1					2.5	2.7	2

Open Elective II (Seventh Semester) **(3-0-0)**

PI1711: Integrated Computational Materials Engineering **Course Outcomes:** After completion of this course,

CO1: Students will know the concepts of sustainable manufacturing and be trained to make an impact assessment to the environment.
CO2: Students will analyze manufacturing processes towards minimization or prevention of air pollution and water pollution.
CO3: Students will develop the skills to create a life cycle assessment model for sustainable manufacturing indices.
CO4: Students will evaluate green co-rating and its benefits.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	0	1	3	3	3		0		3		2
CO2	3	3	3	2	2	3	3	3		2		3		2
CO3	3	3	3	3	3	3	3	3		0		3		3
CO4	3	3	2	1	2	3	3	3		2		3		3
AVG	3	2.75	2.25	1.5	2	3	3	3		1.33		3		2.5

PI1712: Circular Economy

Introduction: Linear Economy and its emergence, Economic and Ecological disadvantages of linear economy, Replacing Linear economy by Circular Economy, Development of Concept of Circular Economy, A differential - Linear Vs Circular Economy

Characteristics: Material recovery, Waste Reduction, reducing negative externalities, Explaining Butterfly diagram, Concept of Loop

Circular design, innovation and Assessment: Zero waste- Waste Management in context of Circular Economy, Circular design, Research and innovation, LCA, Circular Business, Natural Systems, Green Startup Ecosystems and Opportunities on Global and National Level; Green Business models

Case Studies: Business models, Solid Waste Management / Wastewater, Plastics: A case study, EPR: polluters pay principle, Industrial symbiosis/ Eco-parks, Implementation of Circular Economy on Macro Level

Legal and policy framework: Role of governments and networks, Sharing best practices, Universal circular economy policy goals, India and CE strategy

Text Book and Reference books:

1. Shalini Goyal Bhalla, ‘Circular Economy: Emerging Movement’, Invincible Publisher.
2. Peter Lacy, Jessica Long, ‘The Circular Economy Handbook: Realizing the Circular Advantage’, Wesley Spindler Palgrave Macmillan, UK
3. Walter R Stahel Routledge, ‘Waste to Wealth: The Circular Economy A User's Guide’, 1st

Edition, 2019.

4. Jakob Rutqvist, and Peter Lacy, 'Waste to Wealth: The Circular Economy Advantage', 1st Edition, Palgrave Macmillan.
5. Hans Wiesmeth, and Emer, 'The Circular Economy: Concept and Facts', CENN, 2019.
6. Hans Wiesmeth, 'The Circular Economy: Implementation', CENN, 2019.

Course Outcomes: After completion of this course,

CO1: Students will define circular economy and identify its characteristics

CO2: Students will learn concepts of circular design and business models

CO3: Students will apply the concepts of circular economy in various case studies

CO4: Students will appraise the role of governments and legal framework in implementation of circular economy concepts

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	1	1	1		2	3	1	2	1	1	2		
CO2	2	3	2	2		2	3	1	3	2	3	2	2	2
CO3	2	3	3	3	2	2	3	1	3	3	3	2	3	3
CO4	2	3	3	3	2	2	3	1	3	3	2	2	2	2
AVG	1.75	2.5	2.25	2.25	1	2	3	1	2.75	2.25	2.25	2		

PI1713: Industry 4.0

Introduction: Background on previous industrial revolutions, Tools of Industry 4.0, Design principles, Impact on networked economy, New business models, Value proposition and creation, Smart factories, National initiatives and roadmaps of various countries, esp. Digital India and SAMARTH Udyog Bharat 4.0 initiatives by India.

Basic Principles and Technologies of a Smart Factory: Main Components of Industry 4.0: Internet of Things, Artificial Intelligence, Big Data Analytics, Cloud Services, Augmented/Virtual/Mixed Reality, Simulation, Cyber Security, Cyber Physical Systems, Autonomous/Collaborative Robots, 3D Printing, Blockchains Design Principles: Decentralization, Interoperability, Transparency, Virtualization, Modularity, Service Orientation, Basics of Connectivity, Networking and Security.

Industrial Internet of Things: Introduction, Business Model and Reference Architecture, Sensing and Actuation, Processing, Communication and Networking. Communication and networking protocols. Issues and exploring possible solutions.

Hands-on Training on making simple IoT devices: GPS sensors, Camera sensors, and Temperature Sensors.

Artificial Intelligence and Big-Data Analytics in Production: Basics of Machine Learning

Challenges in Application of Industry 4.0: Business issues: Opportunities and Challenges, Strategies, Safety for connected machines and systems, Security and Privacy risks, Lack of digital strategy, visualization, infrastructure, connectivity, skilled workers, legal and financial constraints.

Application of Industry 4.0 in Production Management: Application of Industry 4.0 in Supply Chain Management, Quality Management, Predictive Maintenance, Smart Enterprise Resource Planning (ERP) and Manufacturing Execution Systems (MES), Lean Production, Reconfigurable Manufacturing Systems (RMS), Product Lifecycle Management (PLM).

Application of Industry 4.0 tools in various industries: Applications in Healthcare, Quality Control, Facility Management, Automobile, Oil, Chemical and Pharmaceutical Industries, etc.

Text and Reference Books:

1. Alastair Gilchrist, ‘Industry 4.0: The Industrial Internet of Things’, Publisher: A Press, 2006.
2. Klaus Schwab, ‘The Fourth Industrial Revolution’, World Economic Forum.
3. Christoph Jan Bartodziej, ‘The Concept Industry 4.0: An Empirical Analysis of Technologies and Applications in Production Logistics’, Springer Fachmedien Wiesbaden GmbH, 2017.
4. Alp Ustundag and Emre Cevikcan, ‘Industry 4.0: Managing the Digital Transformation’, Springer International Publishing Switzerland, 2018.
5. Ibrahim Garbie, ‘Sustainability in Manufacturing Enterprises Concepts, Analyses and Assessments for Industry 4.0’, Springer International Publishing Switzerland, 2016.
6. Nikolas E. Karkalos, Angelos P. Markopoulos and J. Paulo Davim, ‘Computational Methods for Application in Industry 4.0’, Publisher: Springer, 2019.

Course Outcomes: After completion of this course,

CO1: Students will learn about the industrial revolution and identify associated technologies and initiatives.

CO2: Students will. apply the concepts of internet of things to create simple IoT devices.

CO3: Students will learn to apply the concepts of data analytics and artificial intelligence in model building.

CO4: Students will apply the concepts of Industry 4.0 in production management problems.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	3		2		1	1		2	3	3
CO2	3	3	3	3	3		2		3			2	3	3
CO3	3	3	3	3	3	3	2		3	2		2	3	3
CO4	3	3	3	3	3	2	2	1	3		2	2	3	3
AVG	3	3	2.75	2.75	3	1.25	2	0.25	2.5	0.75	0.5	2	3	3

PI1714: Warehouse Management

Storage and Packaging Background – Need for Warehouse – Importance of warehouse - Types of Warehouses -Broad functions in a warehouse -warehouse layouts and layout related to functions. Associate warehouse -Its functions -equipment available in associate warehouse, Warehouse Organization Structure -Benefits of Warehousing.

Receiving and Dispatch of Goods in warehouse Various stages involved in receiving goods – Stages involved receipt of goods-Advanced shipment notice (ASN) or invoice items list-Procedure for Arranging of goods on dock for counting and Visual inspection of goods unloaded-Formats for recording of goods unloaded from carriers-Generation of goods receipt note using computer-Put away of Goods-Put away list and its need-Put away of goods into storage locations -storage location codes and its application-Process of put away activity-Procedure to Prepare Warehouse dispatches

Warehouse Activities: Receiving, sorting, loading, unloading, Picking Packing and dispatch, activities and their importance in a warehouse -quality parameters -Quality check-need for quality check-importance of quality check. Procedure to develop Packing list / Dispatch note-Cross docking method -Situations suited for application of cross docking -Information required for coordinating cross docking-Importance of proper packing-Packing materials -Packing machines -Reading labels

Warehouse Management: Warehouse Utilization Management -Study on emerging trends in warehousing sector -DG handling -use of Material Handling Equipment's in a warehouse -Inventory Management of a warehouse -Inbound & Outbound operations of a warehouse and handling of Inbound & Outbound operations. Distribution – Definition – Need for physical distribution – functions of distribution – marketing forces affecting distribution. The distribution concept – system perspective. Channels of distribution: role of marketing channels – channel functions – channel structure –designing distribution channel – choice of distribution channels

Warehouse Safety Rules and Procedures: The safety rules and 'Procedures to be observed in a Warehouse -Hazardous cargo – Procedure for Identification of Hazardous Cargo -safety data sheet-Instructions to handle hazardous cargo -Familiarization with the industry. Health, Safety & Environment -safety Equipment's and their uses -5S Concept on shop floor. Personal protective Equipment's (PPE) and their uses.

Text and Reference Books:

1. Gwynne Richard, 'Warehouse Management– Student Study Guide',
2. Max Muller, 'Essentials of Inventory Management', Harper Collins Publishers.
3. David E. Mulcahy, 'Warehouse Distribution & Operations Handbook',
4. Edward H Frazelle, 'Inventory strategy',
5. V. R. Rangarajan, 'Basics of Warehouse and Inventory Management', 1st Edition,Notion .Press Publisher
6. J. P. Saxena, 'Warehouse Management and Inventory Control', 1st Edition, Vikas Publication House Pvt. Ltd., 2003.
7. Michael Ten Hompel, Thorsten Schmidt, 'Warehouse Management: Automation and Organisation of Warehouse and Order Picking Systems', 1st Edition, Springer Verlag, 2006.

Course Outcomes: After completion of this course,

CO1: Students will classify various types of warehouses in supply chain management.

CO2: Students will discuss the process of receiving and dispatching the goods.

CO3: Students will practice the operations management in warehouses.

CO4: Students will illustrate the aspects of health and safety in warehousing activities.

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	2						1	3	3		3
CO2	3	2	2	2	3	1				1	3	3	2	3
CO3	3	2	2	2	2	2				1	3	3	2	3
CO4	3	2	2	1		3				1	2	1	2	3
AVG	3	2	2	1.75	1.25	1.5				1	2.75	2.5	1.5	3